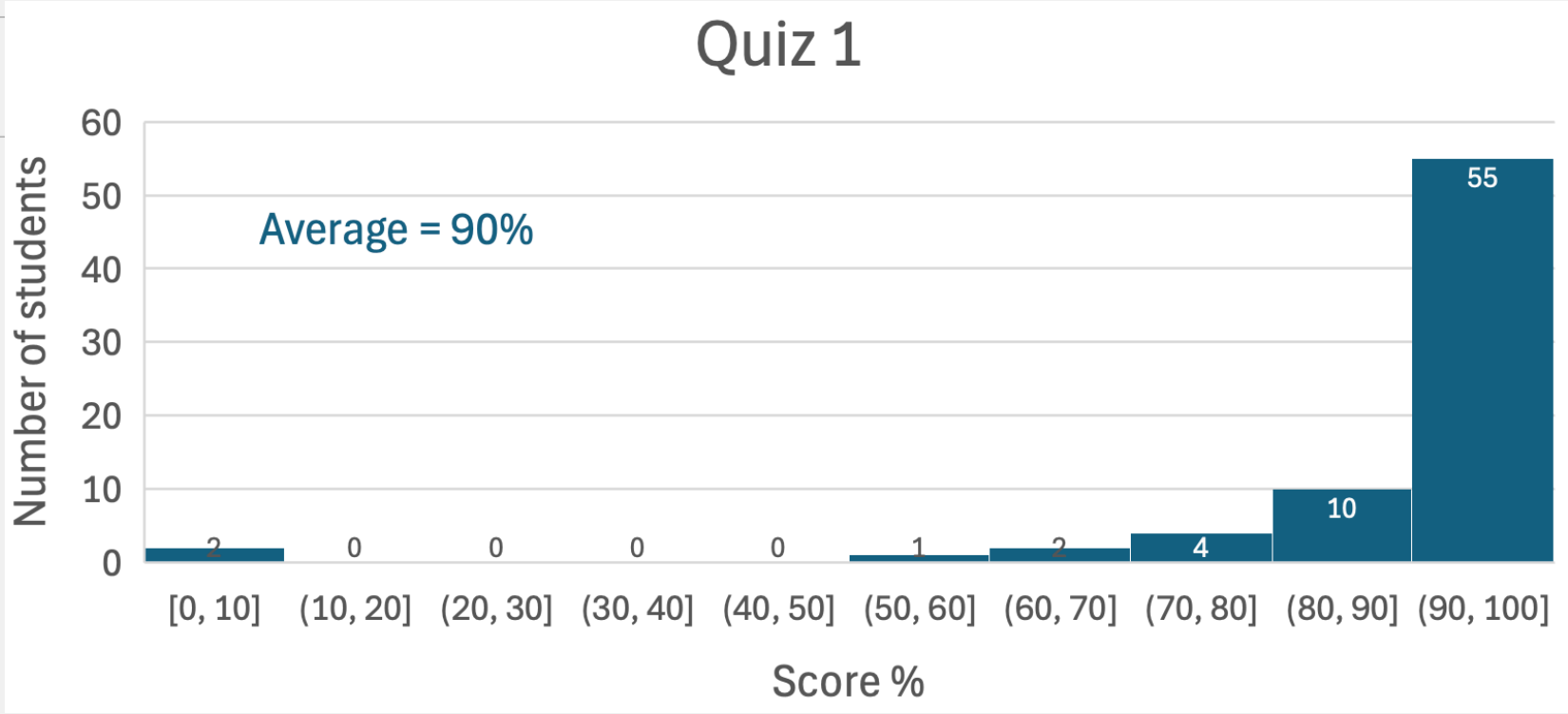
Course Schedule

	MON	TUES	WED	THU	FRI
Course overview and Intro				08/20 First day of classes	
Module 1: The Sun and the Solar Wind		08/25 Review of E&M and Plasma		08/27 The Sun and Solar Wind	
		09/01 The Solar Wind and Heliosphere		09/03 The Solar Effects	9/4 HW1 due (midnight)
Module 2: The Atmosphere		09/08 The Atmosphere	9/9 HW1 revision due (midnight); MOD 1 QUIZ, open at 8am	09/10 The Atmosphere	
		09/15 The Atmosphere		09/17 The Atmosphere	9/18 HW2 due (midnight)
Module 3: The Ionosphere		09/22 The Ionosphere	9/23 HW2 revision due (midnight); MOD 2 QUIZ, open at 8am	09/24 The Ionosphere	
		9/29 The Ionosphere		10/01 The Ionosphere	10/02 HW3 due (midnight)
Module 4: The Magnetosphere		10/06 The Magnetosphere	10/07 HW3 revision due (midnight); MOD 3 QUIZ, open at 8am	10/08 Reading Day	
		10/13 The Magnetosphere		10/15 The Magnetosphere	
Module 5: The Radiation Environment		10/20 The Magnetosphere		10/22 Radiation	10/23 HW4 due (midnight)
		10/27 Radiation	10/28 HW4 revision due (midnight); MOD 4 QUIZ, open at 8am	10/29 Radiation	
Module 6: Micrometeoroids and Orbital Debris (MMOD)		11/03 Radiation		11/05 MMOD	10/06 HW5 due (midnight)
		11/10 MMOD	11/11 HW5 revision due (midnight); MOD 5 QUIZ, open at 8am	11/12 MMOD	
Module 7: Comparative Environments		11/17 Mars-Jupiter		11/19 Mars-Jupiter	11/20 HW6 due (midnight)
Thanksgiving Break	11/23	11/24	11/25	11/26	11/27
		12/01 Mars-Jupiter	12/02 HW6 revision due (midnight); MOD 6 QUIZ, open at 8am	12/03 Review	12/04 HW7 due (midnight) (only graded for completeness) and solutions will be posted soon after
FINAL EXAM WEEK		12/08		**12/10, open at 8am +24 hr; 2.5 hr to complete**	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40
30
20
10
0
Average = 90%
2
0
0
0
0
1
2
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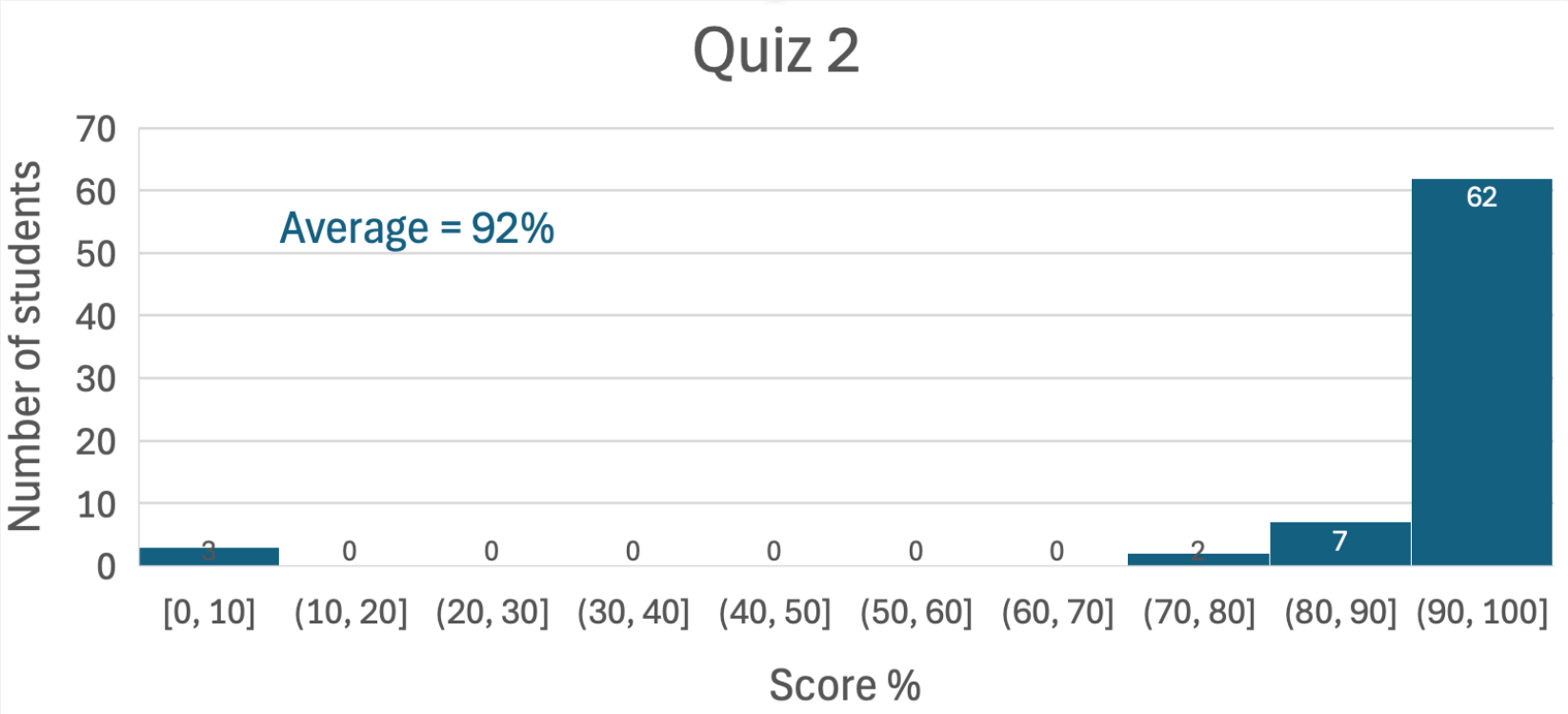
(80, 90]
(90, 100]
Score %


Quiz 2
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(70, 80]
(80, 90]
(90, 100]
Score %

University of Colorado **Boulder**

October 6, 2026

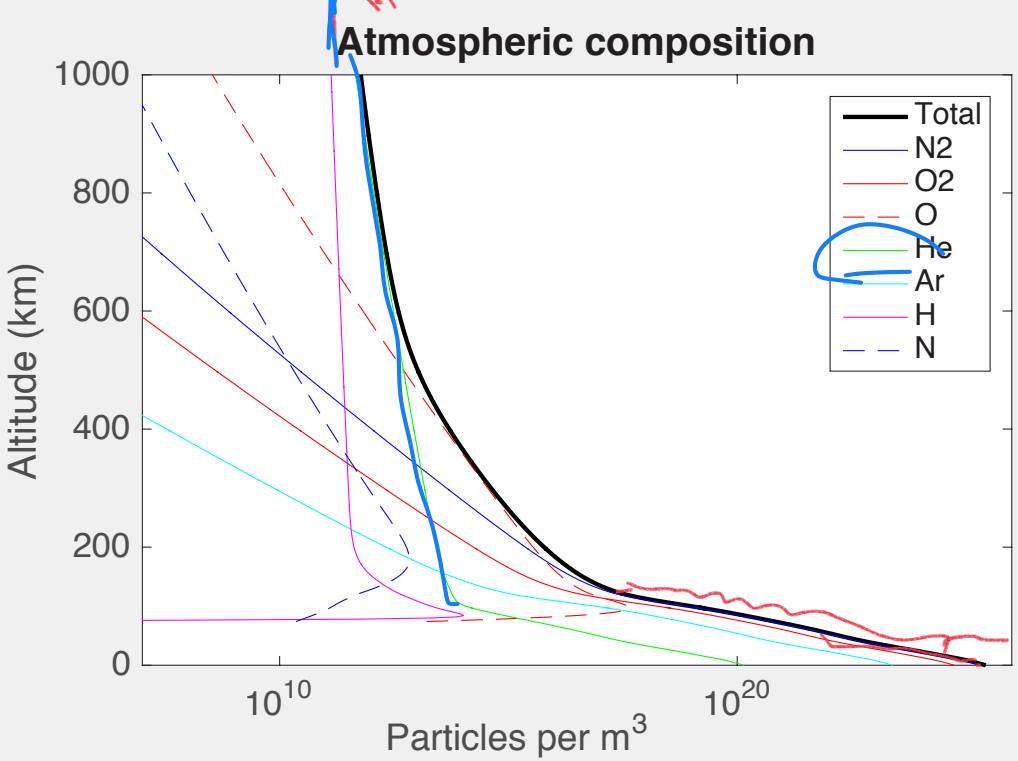
ASEN 5335: Aerospace Environment

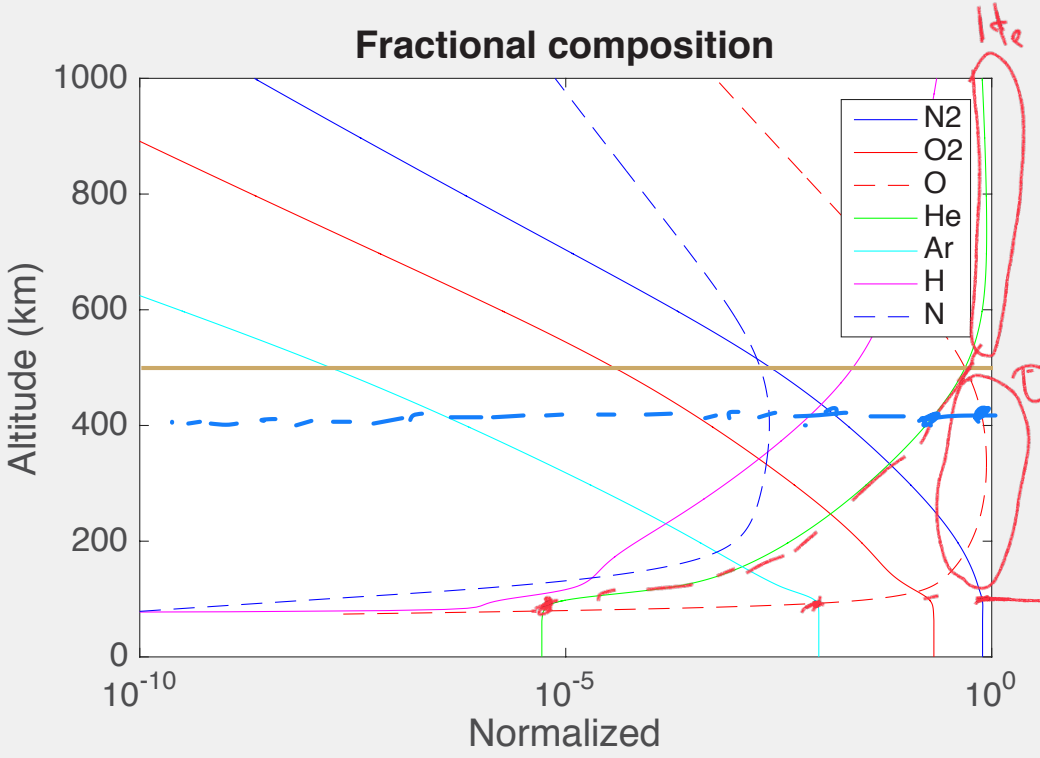
Module 4: The Magnetosphere

Xinlin Li

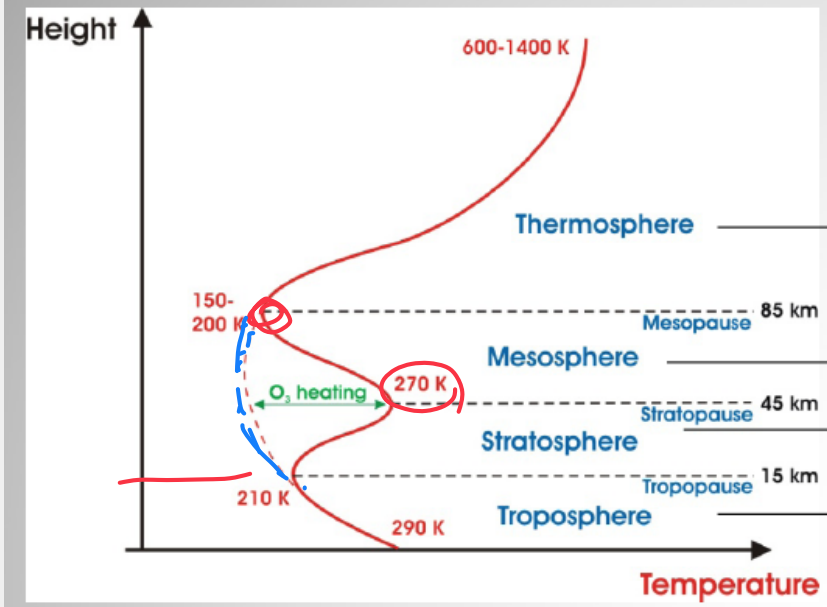
Atmospheric Composition (with altitude)

- **Homosphere**: region below 100 km where composition is constant
  - H = 8 km
- **Heterosphere**: region above 100 km where composition changes with altitude
  - Different scale heights
  - Different regions and effects





Sources and Sinks



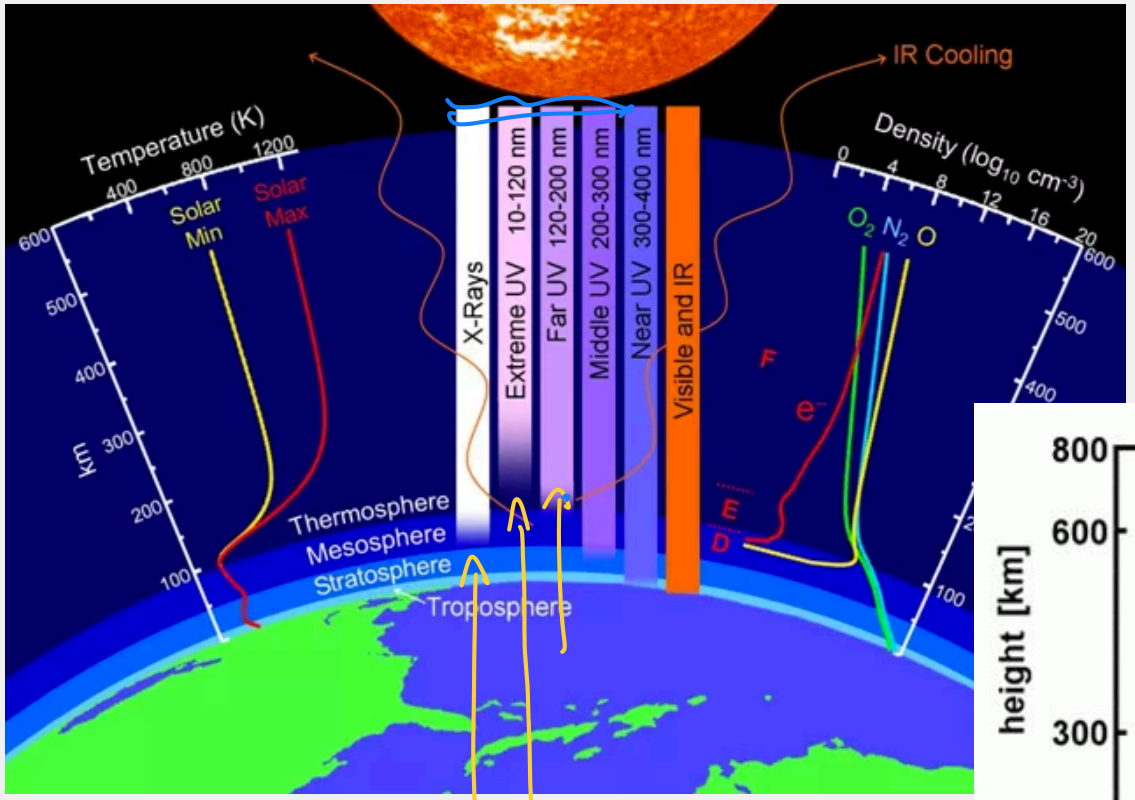
Thermosphere:
- Energy sources:
  - Absorption of EUV (20-100 nm; photoionizing O, O_2, N_2) and UV (120-200 nm), photodissociating O_2), leading to chemical reactions and particle collisions, liberating energy
  - Joule heating by auroral electrical currents
  - Particle precipitation from the magnetosphere
  - Dissipation of upward propagating waves (tides, planetary waves, gravity waves)
- Energy sinks:
  - Thermal conduction into the mesosphere, where energy is radiated by CO_2, O_3 and H_2O
  - IR radiation by CO_2 NO, O

Mesosphere:
- Energy sources:
  - Some UV absorption by O_3 (lower heights)
  - Heat transport down from thermosphere (minor, upper heights only)
  - Chemical heating
- Energy sinks:
  - IR radiation by CO_2 H_2O, OH

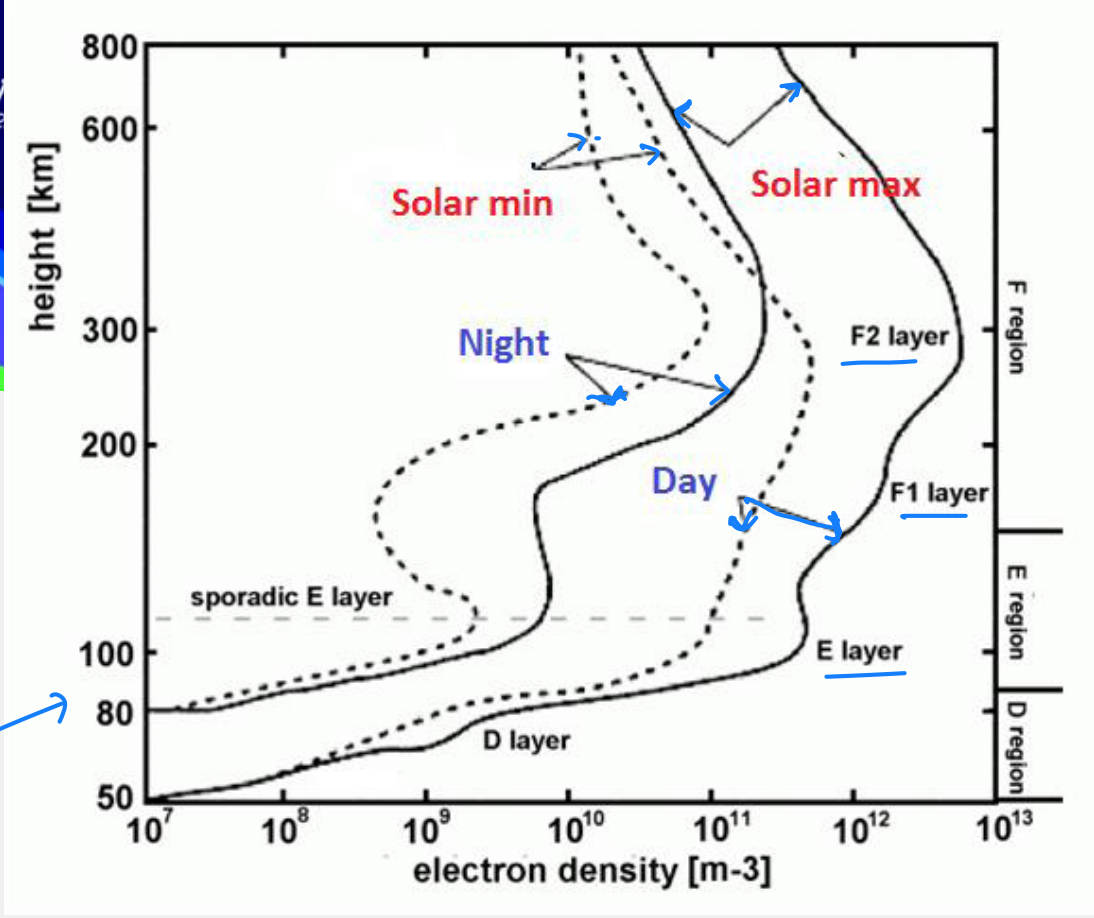
Stratosphere:
- Energy sources:
  - Strong absorption of UV by ozone (causing stratopause temperature peak)
- Energy sinks:
  - IR radiation by O_3 , CO_2 , H_2O

Troposphere:
- Energy sources:
  - Planetary surface absorption (IR, visible), convection & conduction to atmosphere
  - Atmospheric absorption of terrestrial and solar IR
  - Latent heat release by H_2O
- Energy sinks:
  - IR radiation
  - Evaporation of H_2O

Earth's Ionosphere



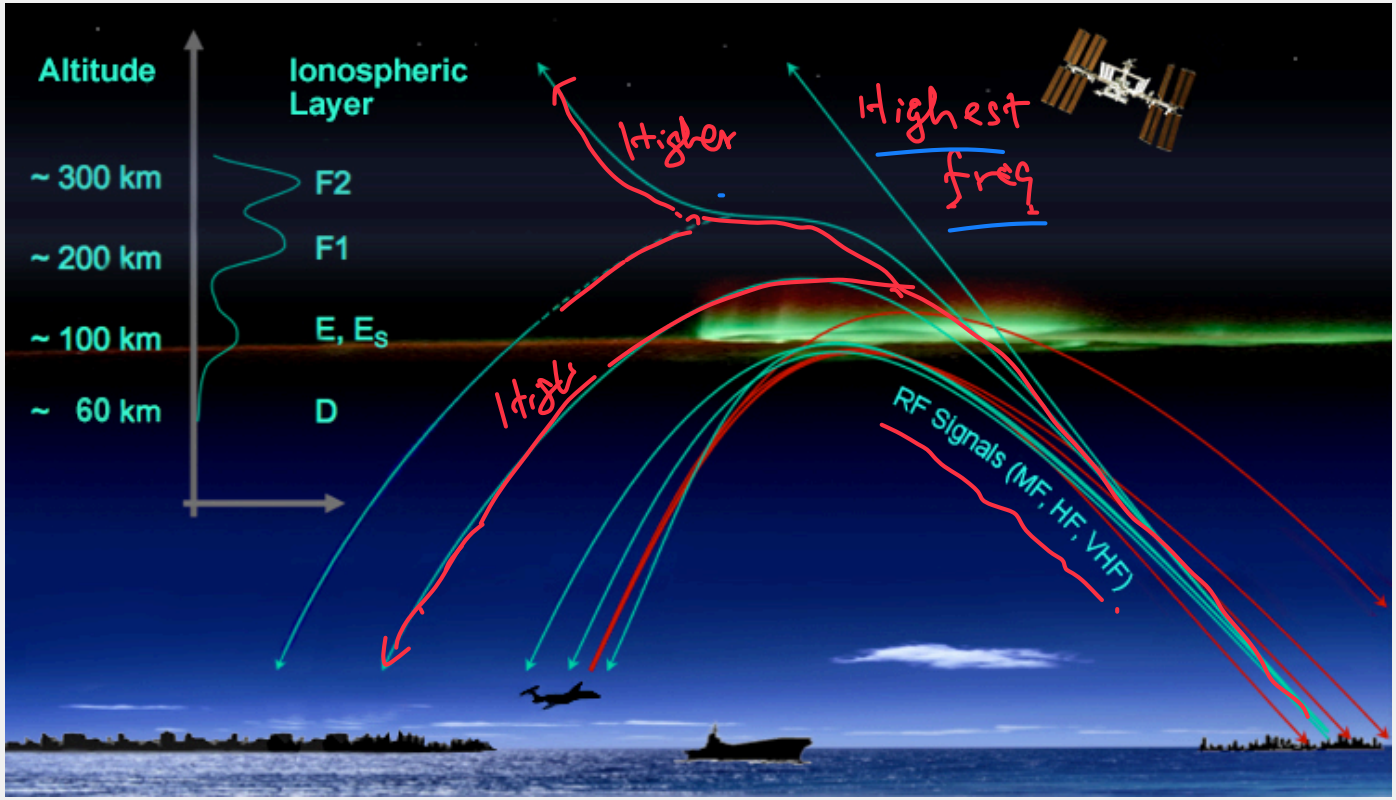
- Ionosphere altitudes and layers have a lot to do with where solar radiation is absorbed!



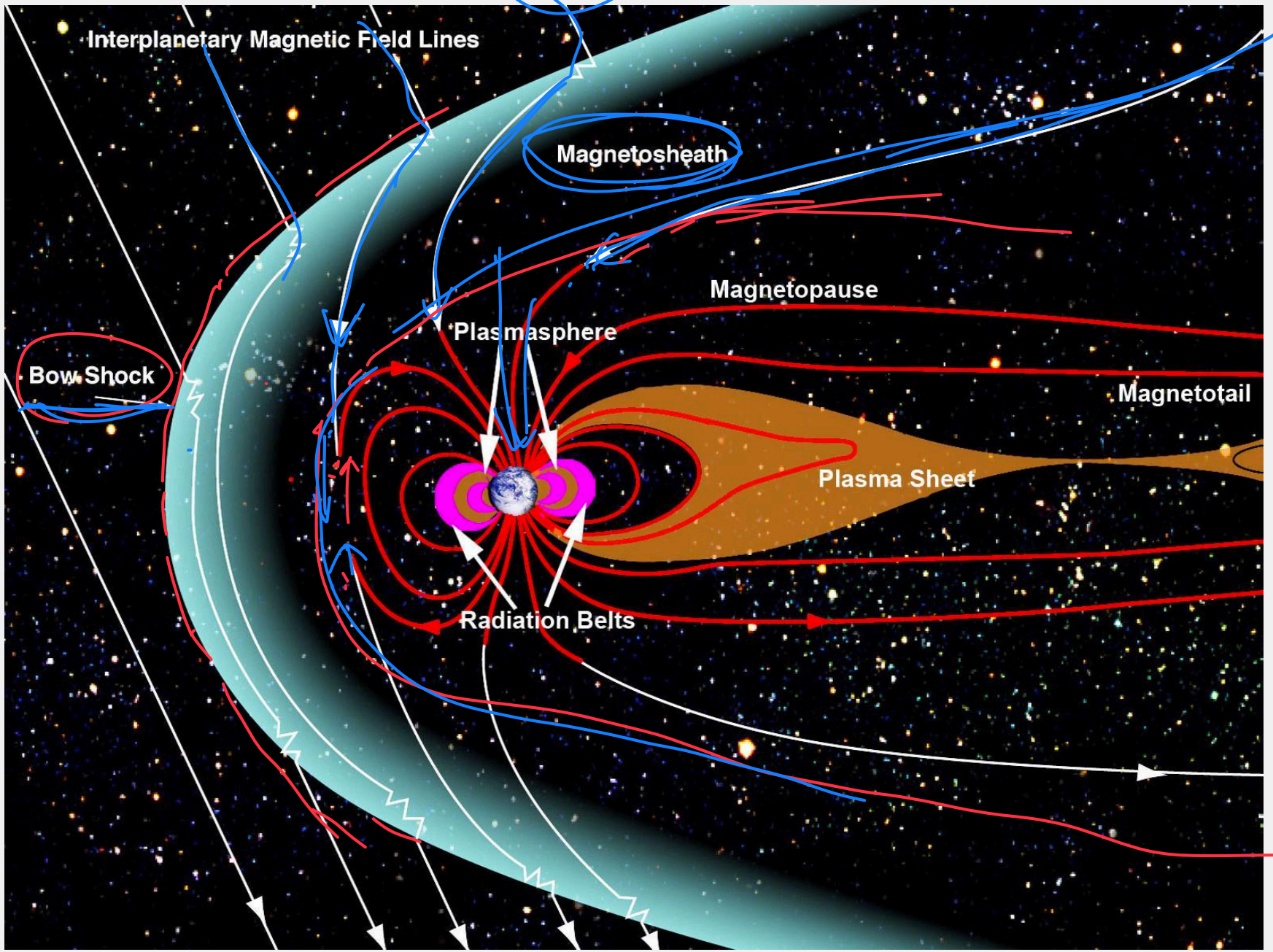
Reflection of EM Waves

- Plasma frequency ω_p directly related to electron density
- Radio waves above ω_p pass through the ionosphere (with some loss); electrons cannot respond fast enough (but still have some oscillations)
- Radio waves below ω_p are reflected; electrons are "shaken" and re-radiate

- **Implications:**
  - must use frequencies above ω_p to communicate to satellites
  - Can communicate over-the-horizon with frequency near / below ω_p



The Magnetosphere



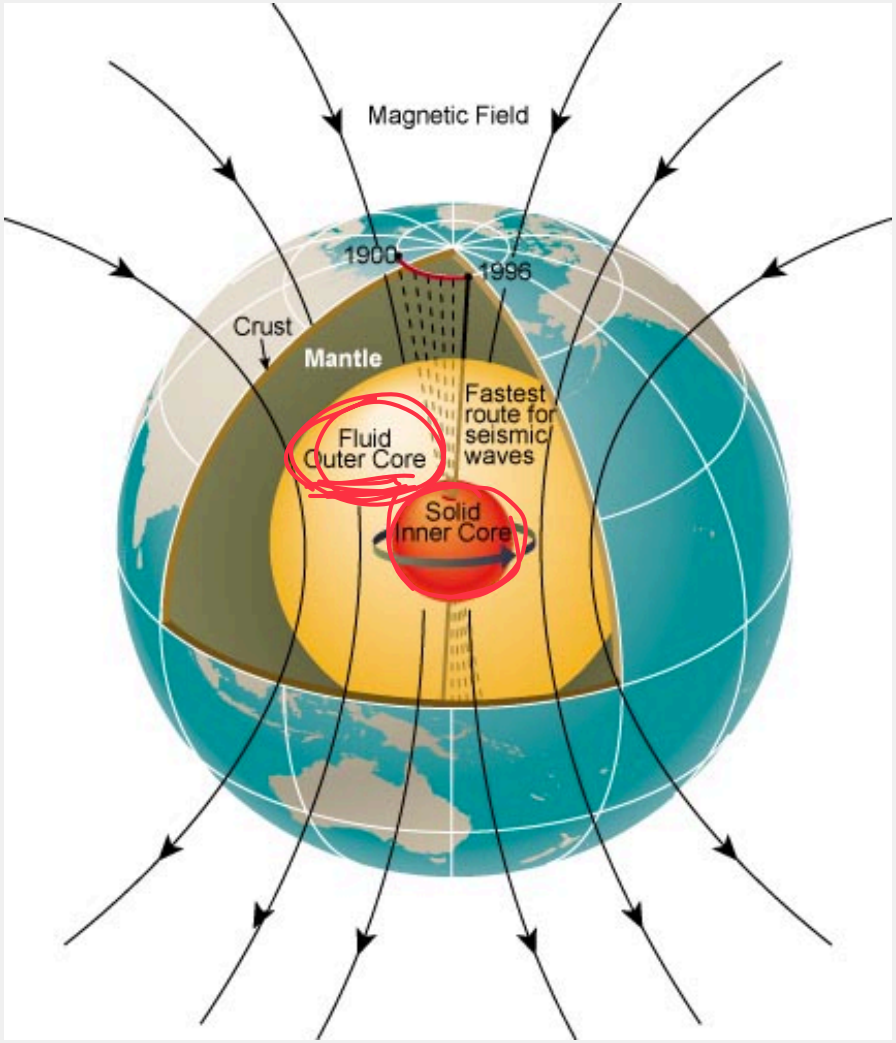
Goals of This Module

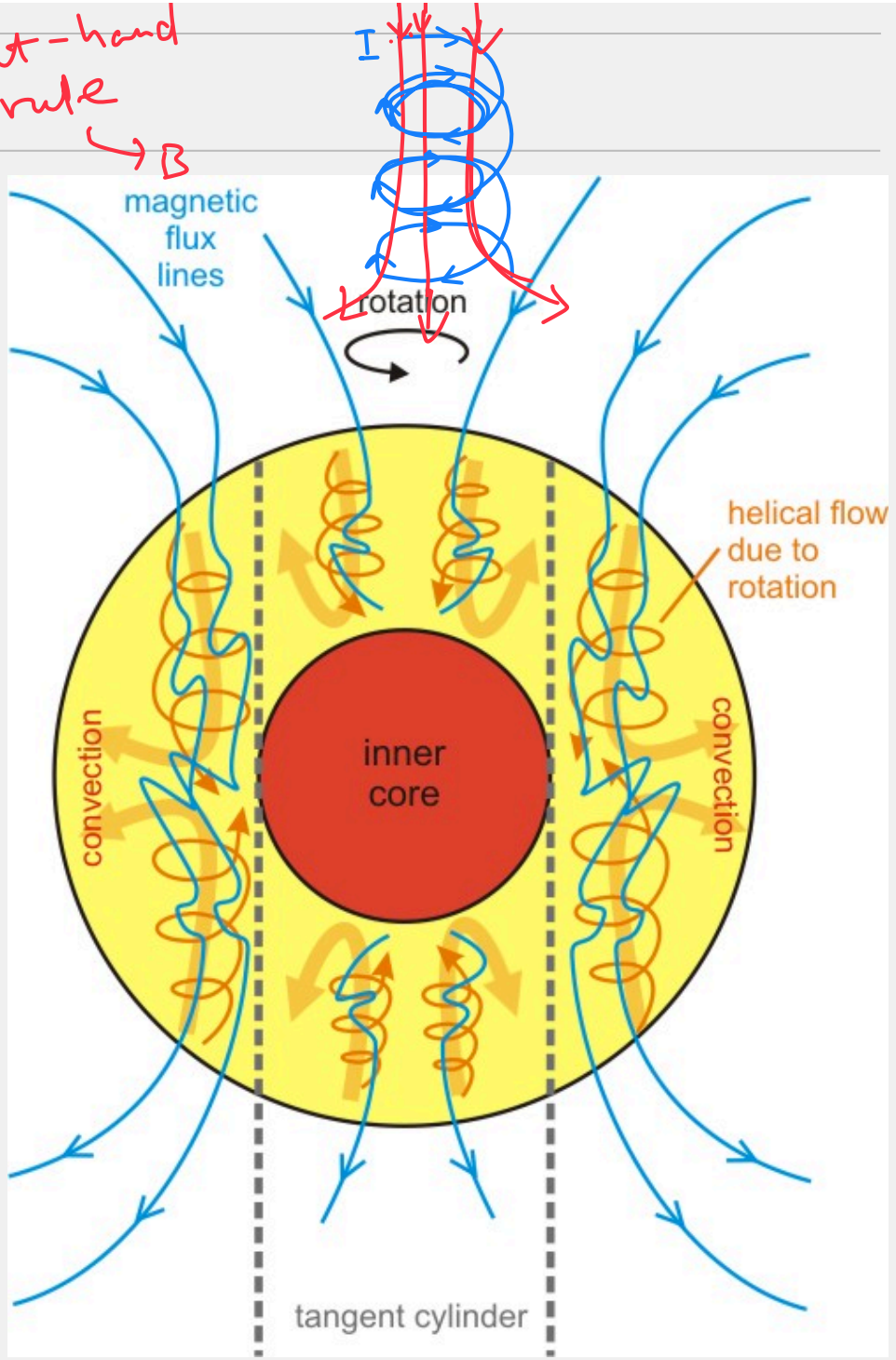
- **Understand basic physics of the Earth's Magnetic Field and Magnetosphere**
  - Origin; surface field; dipole field; extended magnetosphere
  - Variations: solar wind stretching, currents, plasma forcing
  - Particle motion in magnetic fields; current systems in magnetosphere
- **Understand effects of the Magnetosphere on Spacecraft and technology**
  - Navigation
  - Protection from SEPs
  - Geomagnetic storms, GICs
- **Learn about Magnetosphere space missions: Swarm and MMS**

Magnetic Field origin

I : Right-hand rule → B

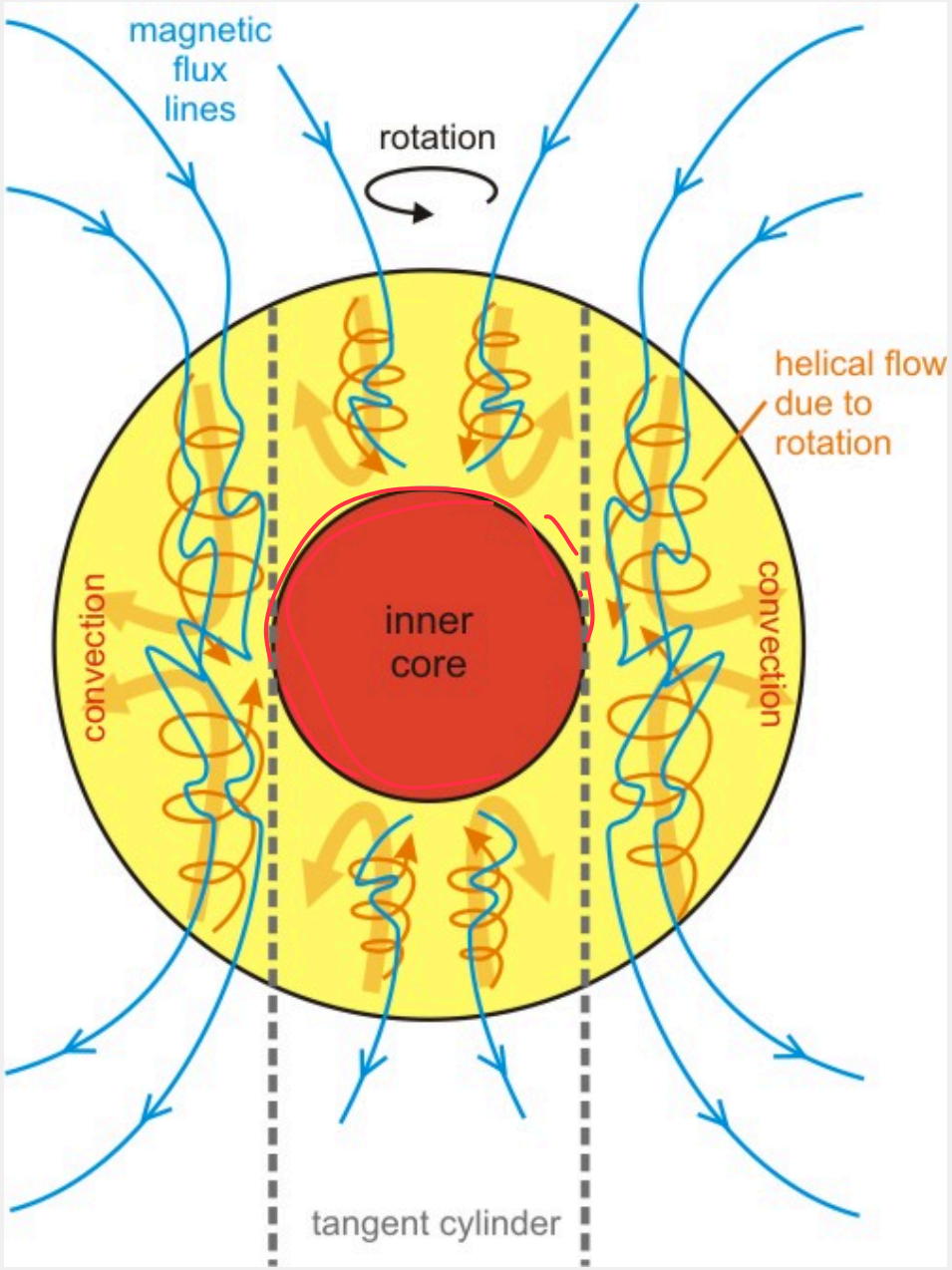
- Just like the sun, B-field originates from rotating, conducting fluid



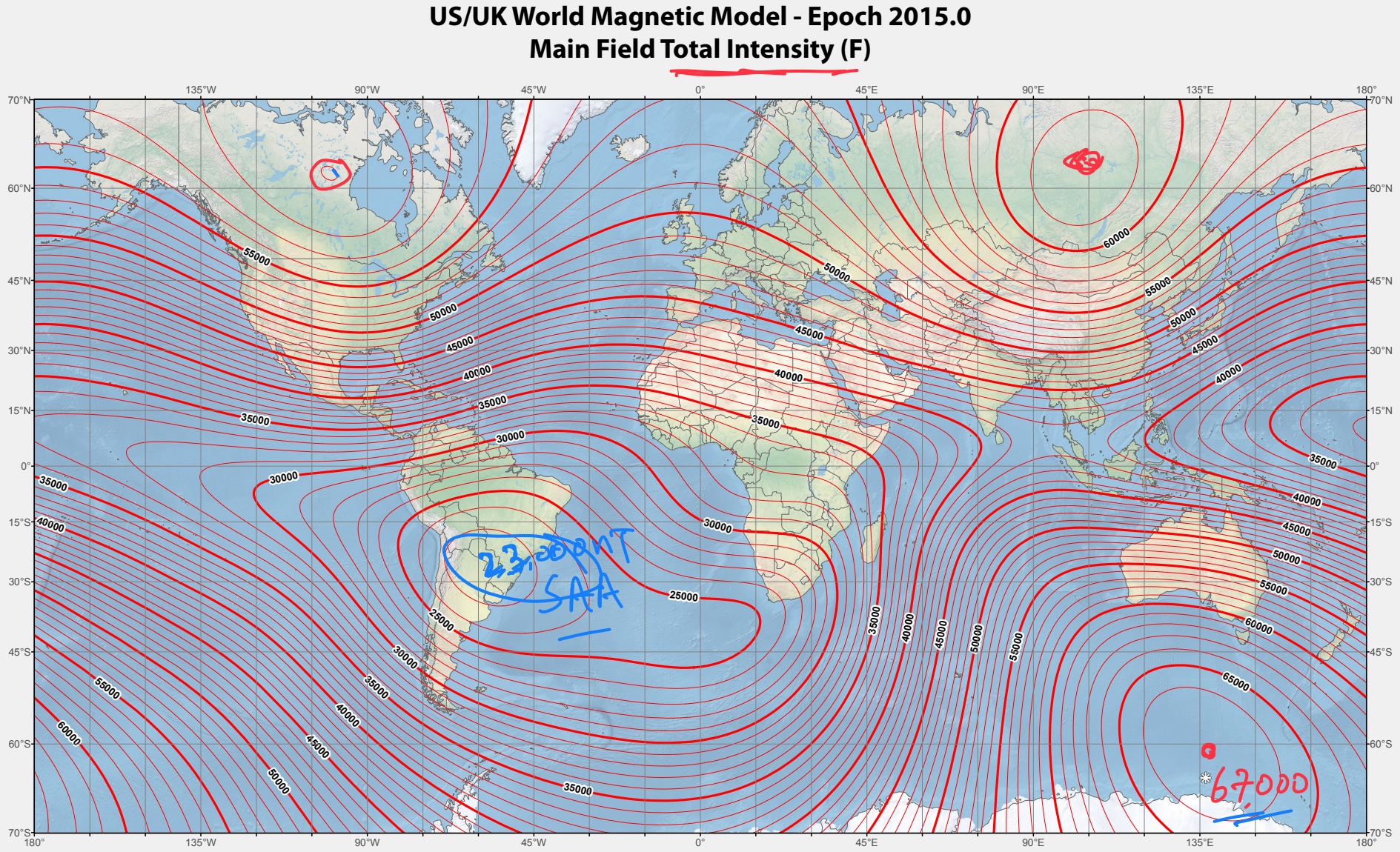


Some interesting factoids

- Inner core is too hot to hold a magnetic field: Curie temperature
  - **Inner core is solid**, crystalline iron
  - Only directly confirmed in 2005
    - Some recent news!
  - Approximately the size of the moon (1/3.67)*Re
  - P = 3.5 x 10^6 x surface pressure; T ~ 6000 C
- **Outer core is liquid** iron/nickel
  - T ~ 4500–6000 C
- Average surface density is 10 g/cm^3; average Earth density is 3 g/cm^3, so the core must be metal
- Measured by seismic waves



Surface Field

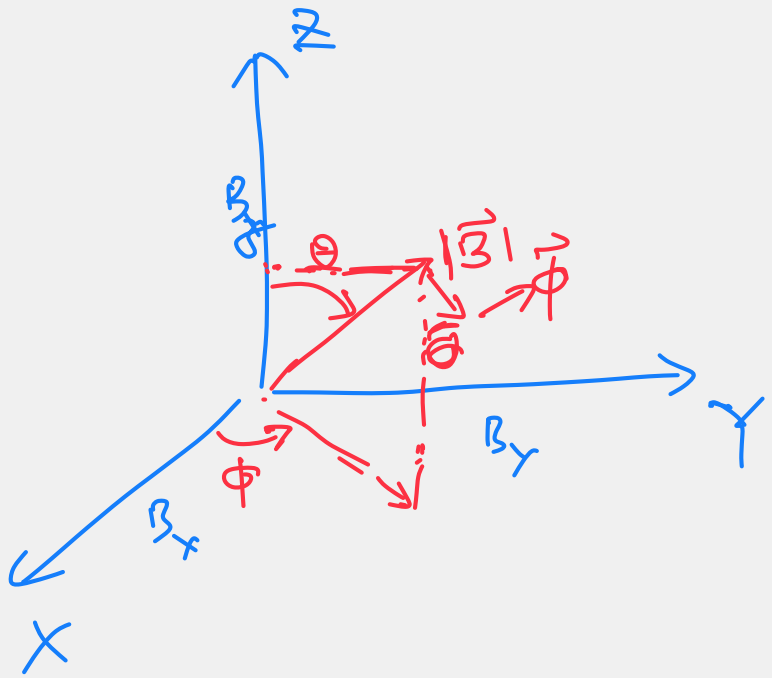


Main Field Total Intensity (F)
Contour interval: 1000 nT.
Mercator Projection.
: Position of dip poles

Map developed by NOAA/NGDC & CIRES
http://ngdc.noaa.gov/geomag/WMM
Map reviewed by NGA and BGS
Published December 2014

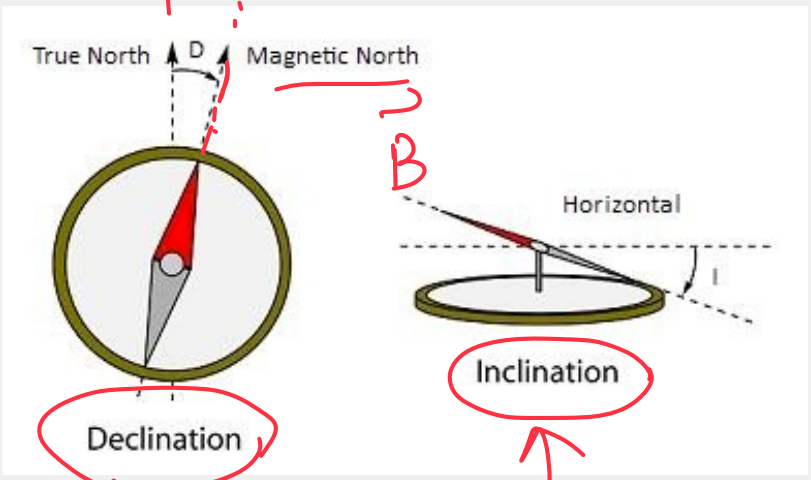
Units for B-field

- Recall magnetic flux density ***B*** is in units of Tesla (T, or nT)
- Magnetic field ***H*** is in units of Amperes/meter (A/m)
- $\boldsymbol{B} = \mu \boldsymbol{H}$
- Both are **vector** fields, with three orthogonal components (B_x, B_y, B_z)
- μ is magnetic permeability; μ_0 is permeability of free space = 1.257 x 10^{-6} H/m
- related to other constants…



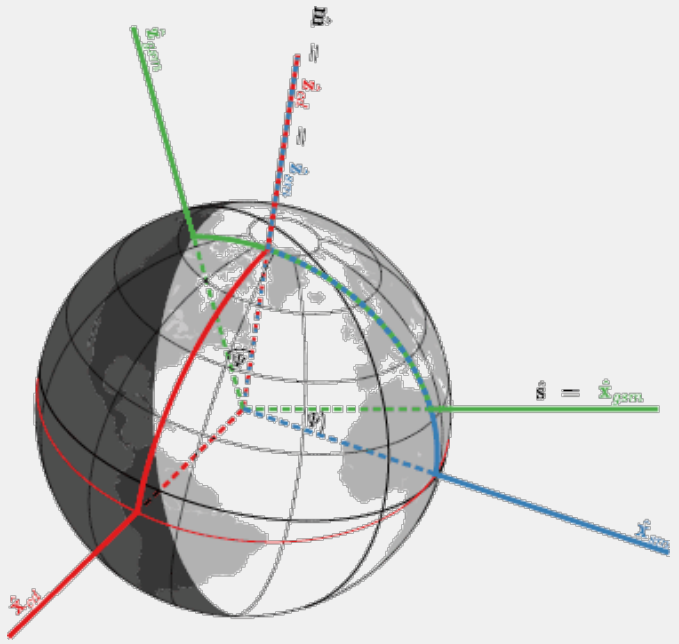
Coordinates!

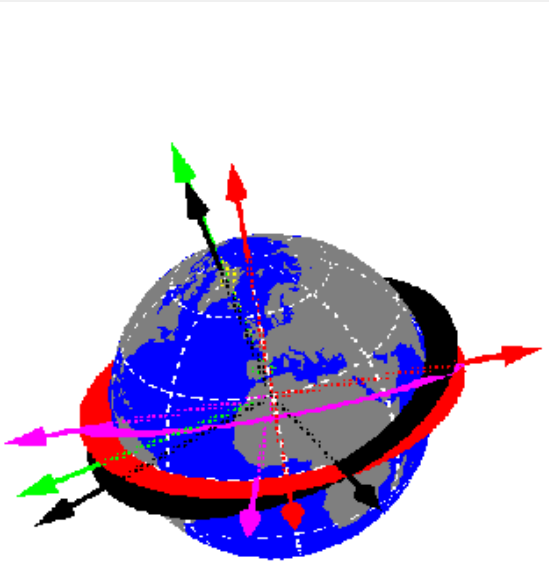
- Normally any vector field has x, y, and z components (B_x, B_y, B_z)
  - Or spherical coordinates: B_r, B_Θ, B_Φ
- On surface of the Earth, it is natural to use the magnitude of B (F), the inclination angle (I), and the declination angle (D)
  - These also form a complete, orthogonal coordinate system
  - Could also use B_V, B_H, D
- In space, this is less natural, so we tend to go back to B_x, B_y, B_z



More on Coordinate Systems

- Geographic
- Geodetic
- Geocentric Equatorial Inertial (GEI)
- Geomagnetic
- Geocentric Solar Ecliptic (GSE)
- Geocentric Solar Equatorial (GSEQ)
- Geocentric Solar Magnetospheric (GSM)
- Solar Magnetic (SM)
- Spacecraft
- Transformation matrices map from one coordinate system to another
- https://www.spenvis.oma.be/help/background/coortran/coortran.html
- https://link.springer.com/article/10.1007/s11214-016-0275-y





Axes

	X	Y	Z
GEO			
GEI			
MAG			
SM			
GSM			

Coordinates

Geodetic:

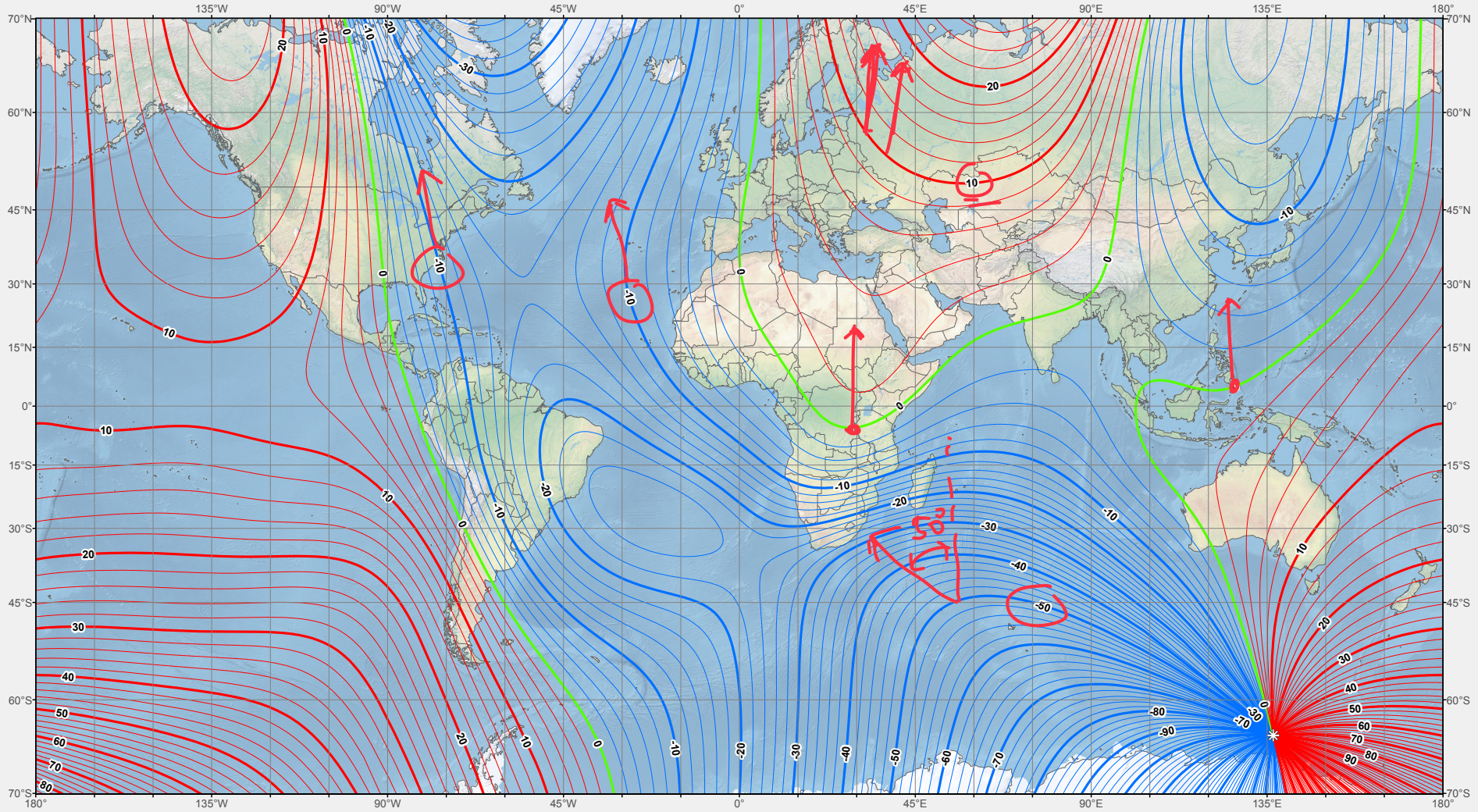
0.1504 50.7978 4.3592

Radius: 6365.5119

	latitude	longitude
GEO	50.6092	4.3592
GEI	50.6092	104.8044
MAG	51.9947	89.1577
SM	51.9947	−167.9438
GSM	26.5415	−171.7345
GEI_{J2000}	50.6120	104.8169

Declination Map

US/UK World Magnetic Model - Epoch 2015.0
Main Field Declination (D)

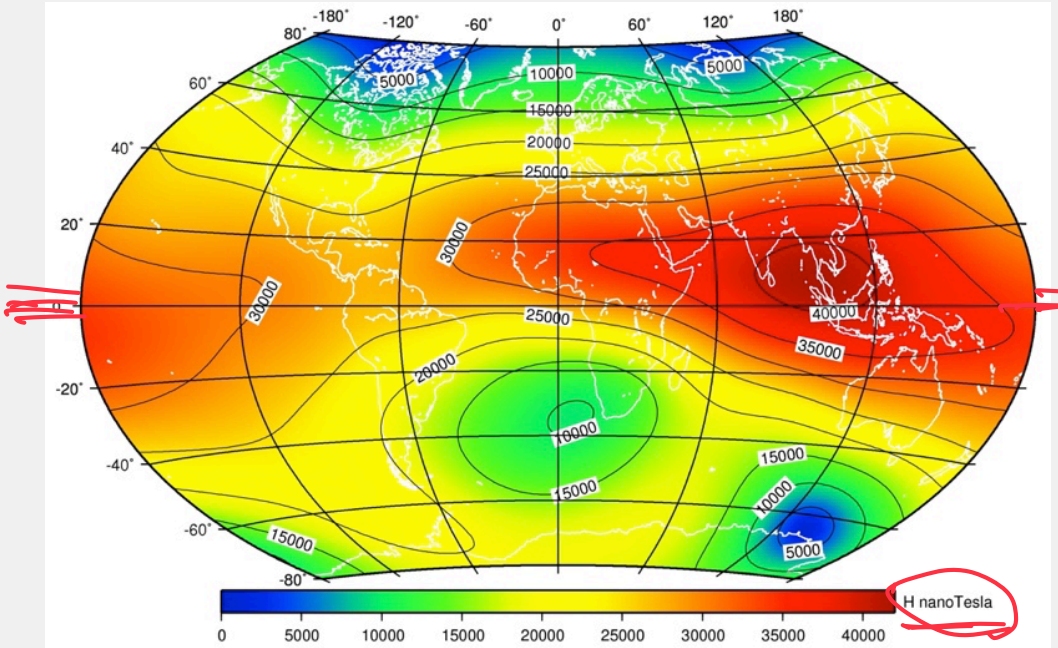


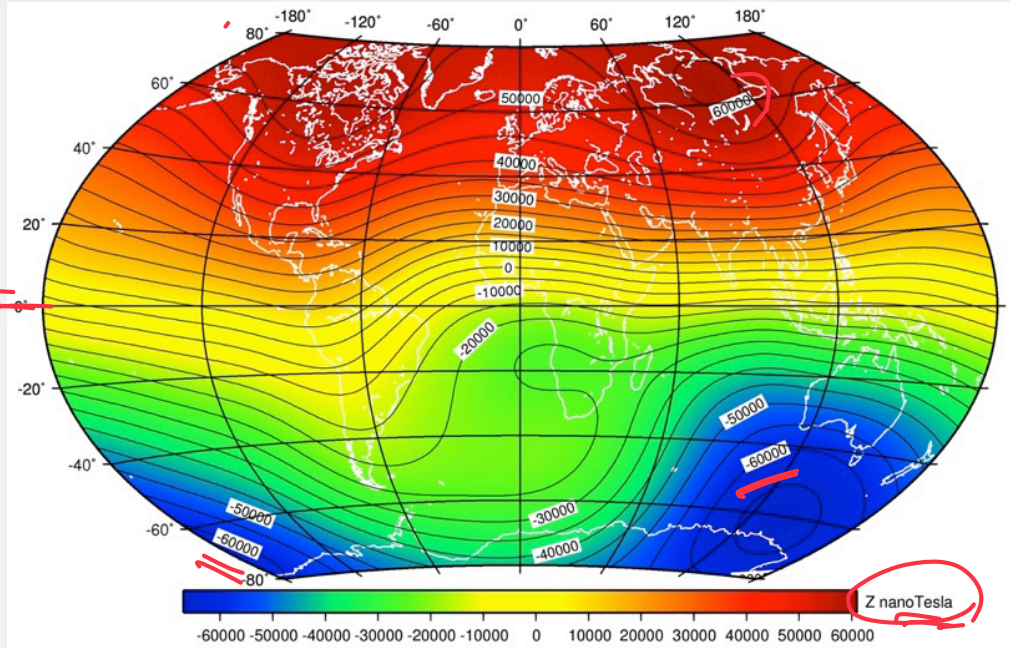
Main field declination (D)
Contour interval: 2 degrees, red contours positive (east); blue negative (west); green (agonic) zero line.
Mercator Projection.
: Position of dip poles

Map developed by NOAA/NGDC & CIRES
http://ngdc.noaa.gov/geomag/WMM
Map reviewed by NGA and BGS
Published December 2014

Another Coordinate Option & IGRF

- Vertical (Z) and Horizontal (H) components, + declination (D)
- More natural coordinates for space applications

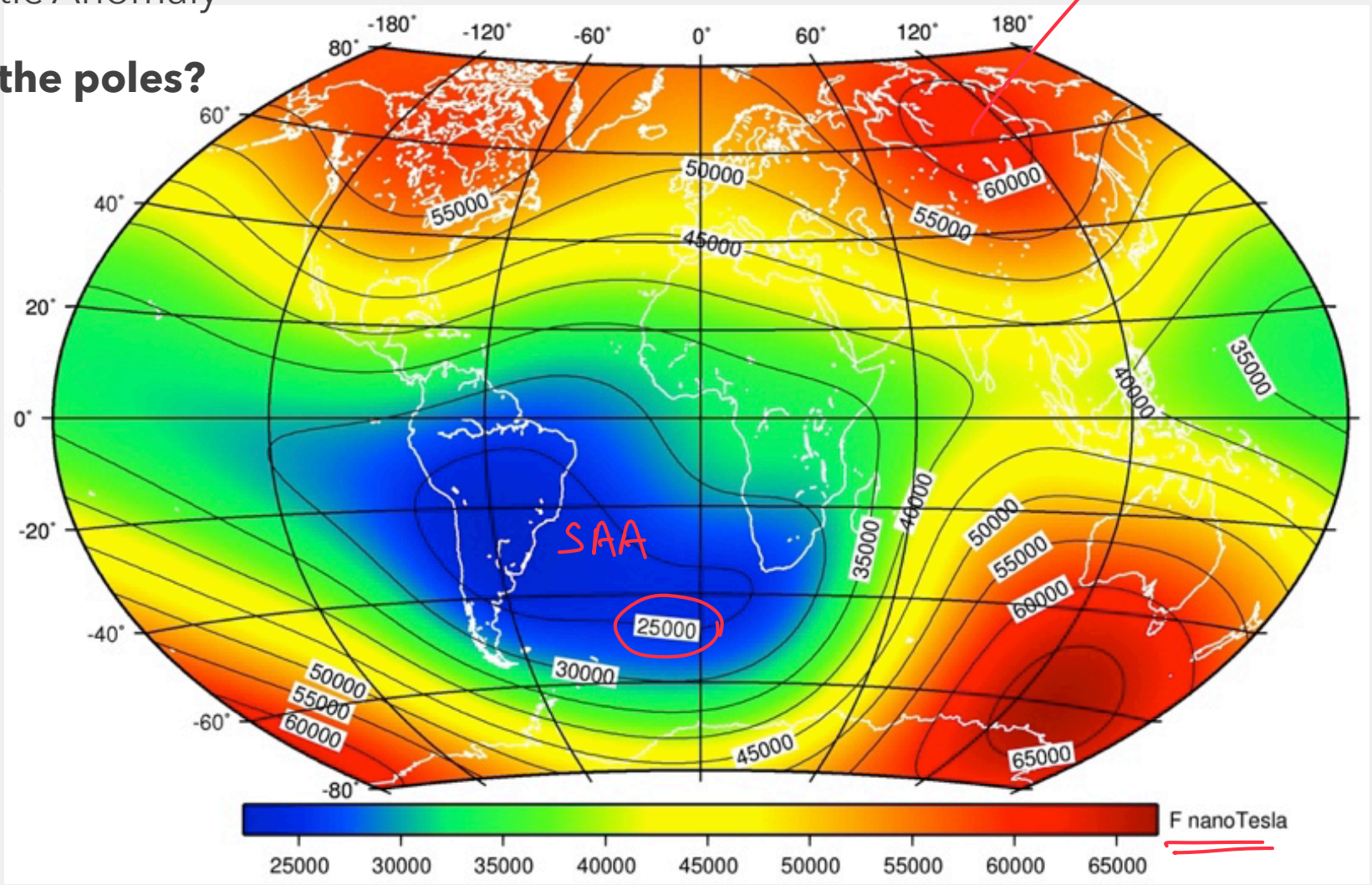




IGRF Total Field

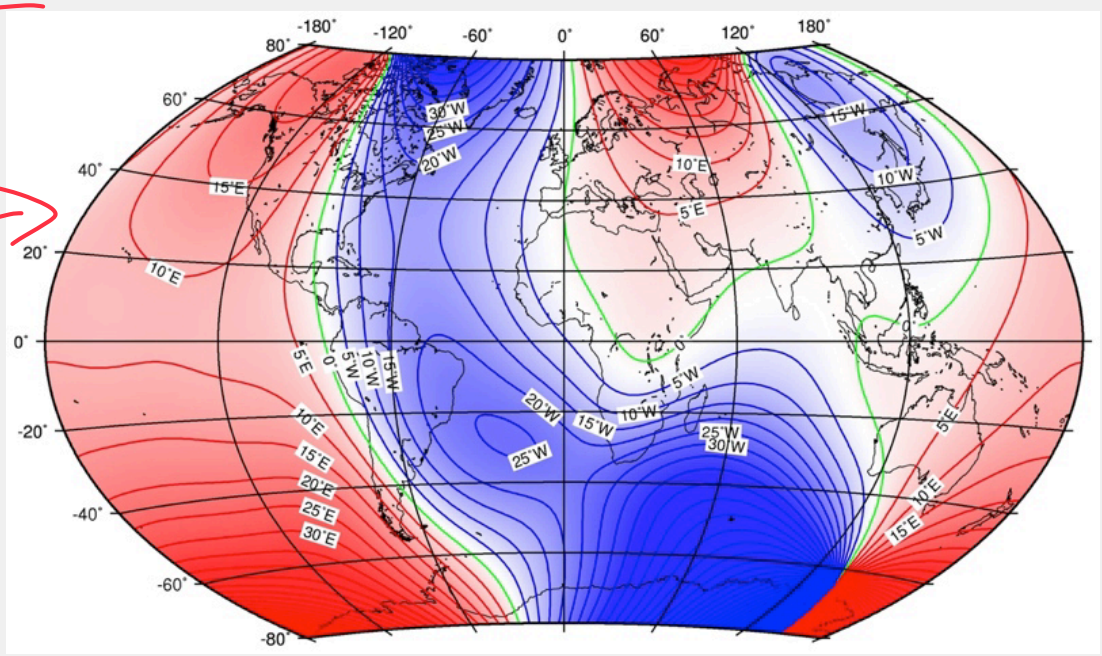
- Total field (F) given by $F^2 = H^2 + Z^2$
- South Atlantic Anomaly
- **Where are the poles?**

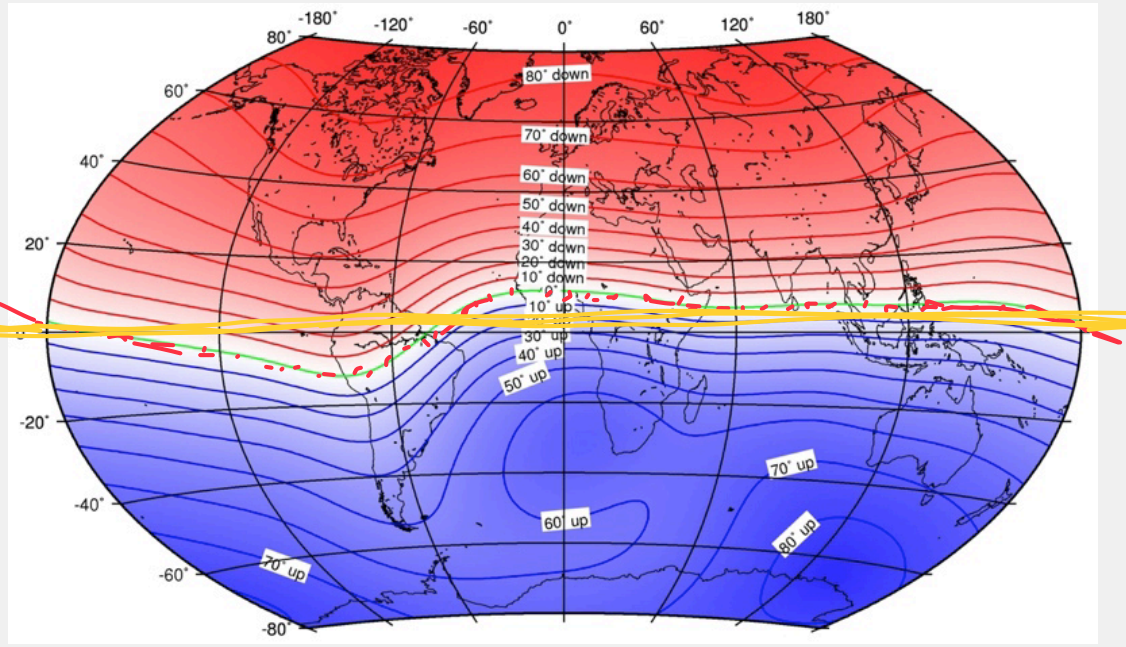
completely vertical



Declination and Inclination

- Declination completes coordinate system
- Inclination determined from H and Z components
- Inclination = 0 defines **magnetic equator**





mag. equator

The Dipole Field Model

$\vec{m} = N \cdot I \cdot A \cdot \vec{n}$ — current, area, unit vector, number of turn of core

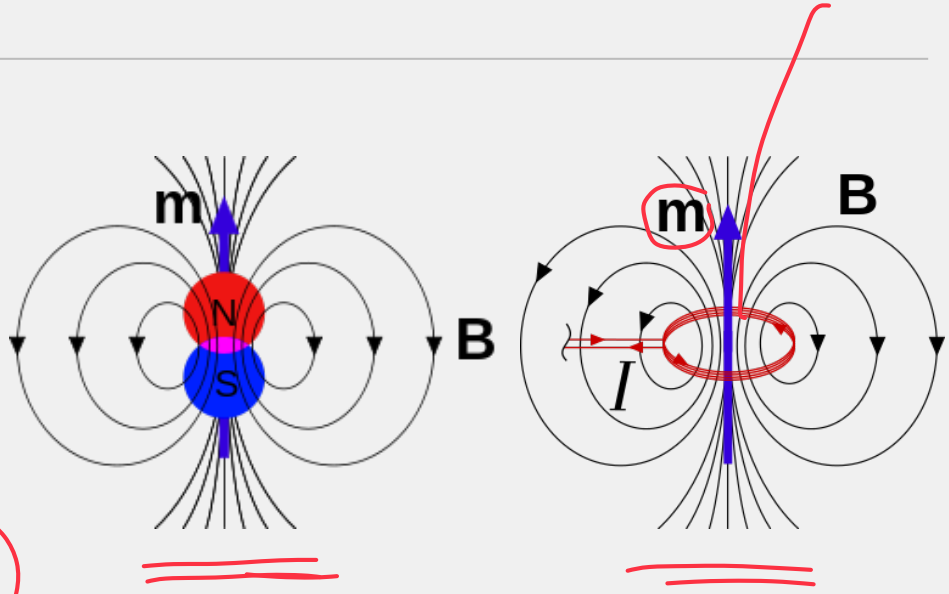
$\nabla \cdot \vec{B} = 0$

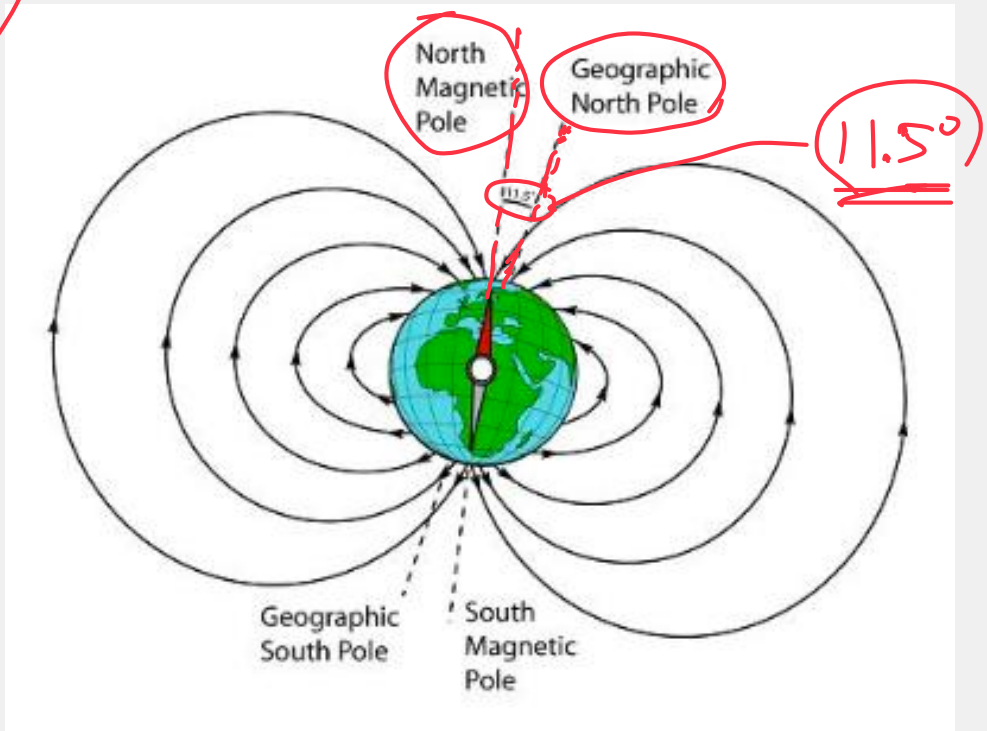
$\nabla \times \vec{H} = \vec{J} + \varepsilon \frac{d\vec{E}}{dt}$

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi}\left(\frac{3\vec{r}\,(\vec{m}\cdot\vec{r})}{r^5} - \frac{\vec{m}}{r^3}\right)$$

$|\vec{B}| \propto \frac{1}{r^3}$

B_r, B_θ, $B_\phi \rightarrow 0$

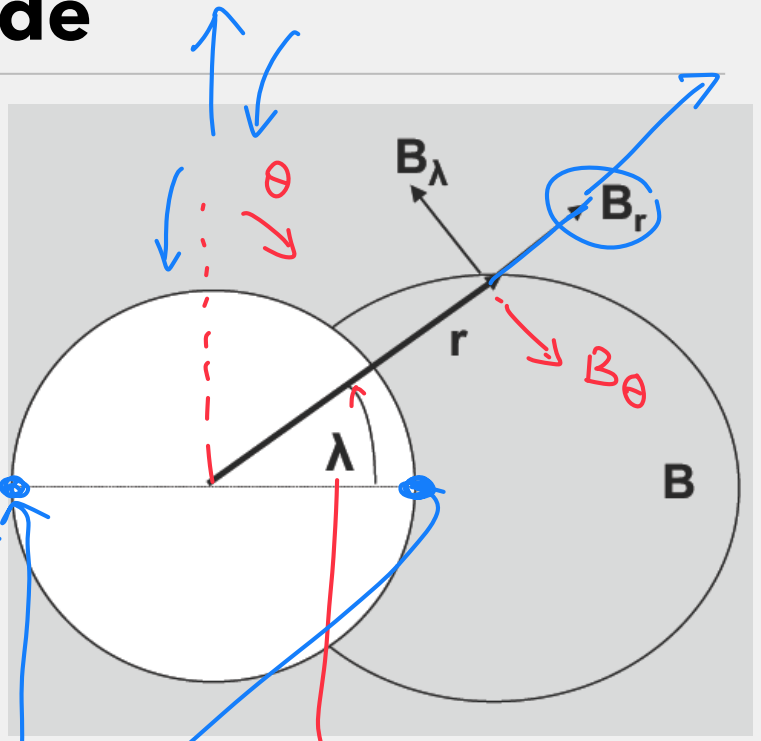




B-field components and magnitude

$$\begin{cases} B_r = -2B_0\left(\frac{R_E}{r}\right)^3 \sin\lambda \\ B_\lambda = B_0\left(\frac{R_E}{r}\right)^3 \cos\lambda \\ B_\phi = 0 \end{cases}$$

$\theta = 90 - \lambda$



$$\frac{M_0 m}{4\pi} = B_0 R_E^3$$

at equator at surface of Earth ≈ 30,000 nT ~ 0.31 G

mag. latitude

at equator: $\lambda = 0$, $r = R_E$

$B_r = 0$, $B_\lambda = B_0 \simeq 31{,}000$ nT

at northpole: $\lambda = 90°$, $r = R_E$

$B_r = -2B_0$, $B_\lambda = 0$

↳ mag. field at poles is a factor of "2" of mag. field at the equator

at south pole:

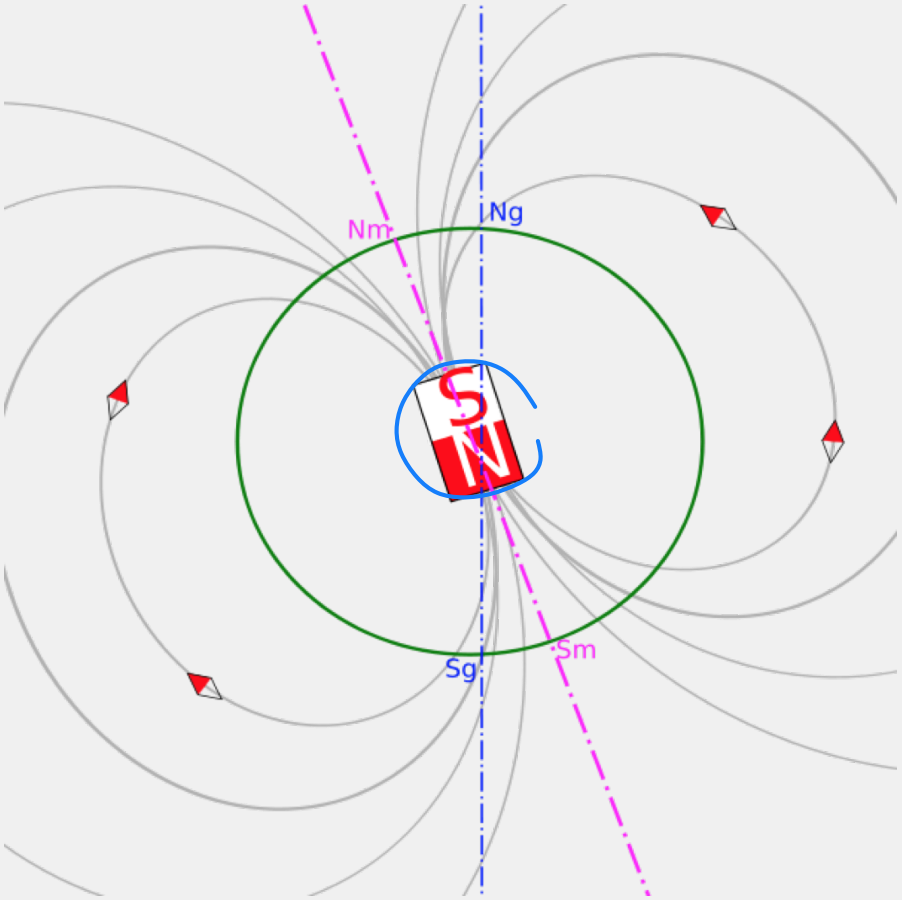
$B_r = 2B_0$

$$|B| = (B_r^2 + B_\lambda^2)^{1/2} = B_0\left(\frac{R_E}{r}\right)^3 \sqrt{1 + 3\sin^2\lambda}$$

centered dipole

Tilted, Shifted Dipole…

- The dipole is **tilted** 11.5 degrees with respect to Earth's spin axis, towards 70 W longitude (Canada)
- and **shifted** towards Russia (away from SAA)
  - Leads to South Atlantic Anomaly
- **With these two corrections, the dipole model has an error about 10% compared to the actual field**



What is a field line?

- The field is tangential to the magnetic field line.
- Field strength is proportional to the line density.
- Field lines cannot cross. $\nabla \cdot \vec{B} = 0$
- Field lines are continuous loops.
- Mathematically: for a dipolar field

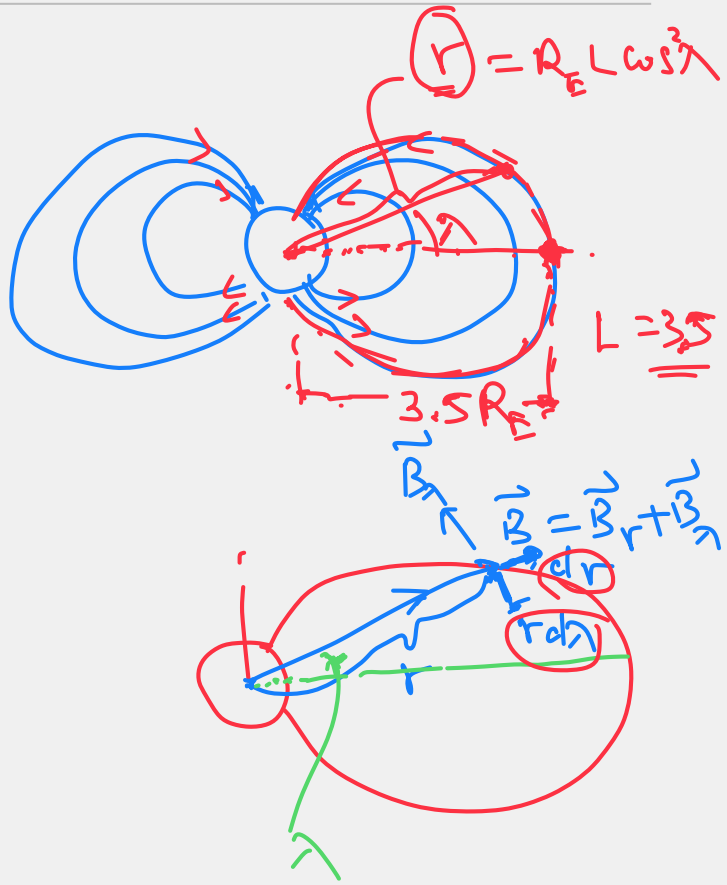
$$\frac{dr}{r d\lambda} = \frac{B_r}{B_\lambda}$$

$$\frac{r d\lambda}{dr} = \frac{1}{2 \tan\lambda}$$

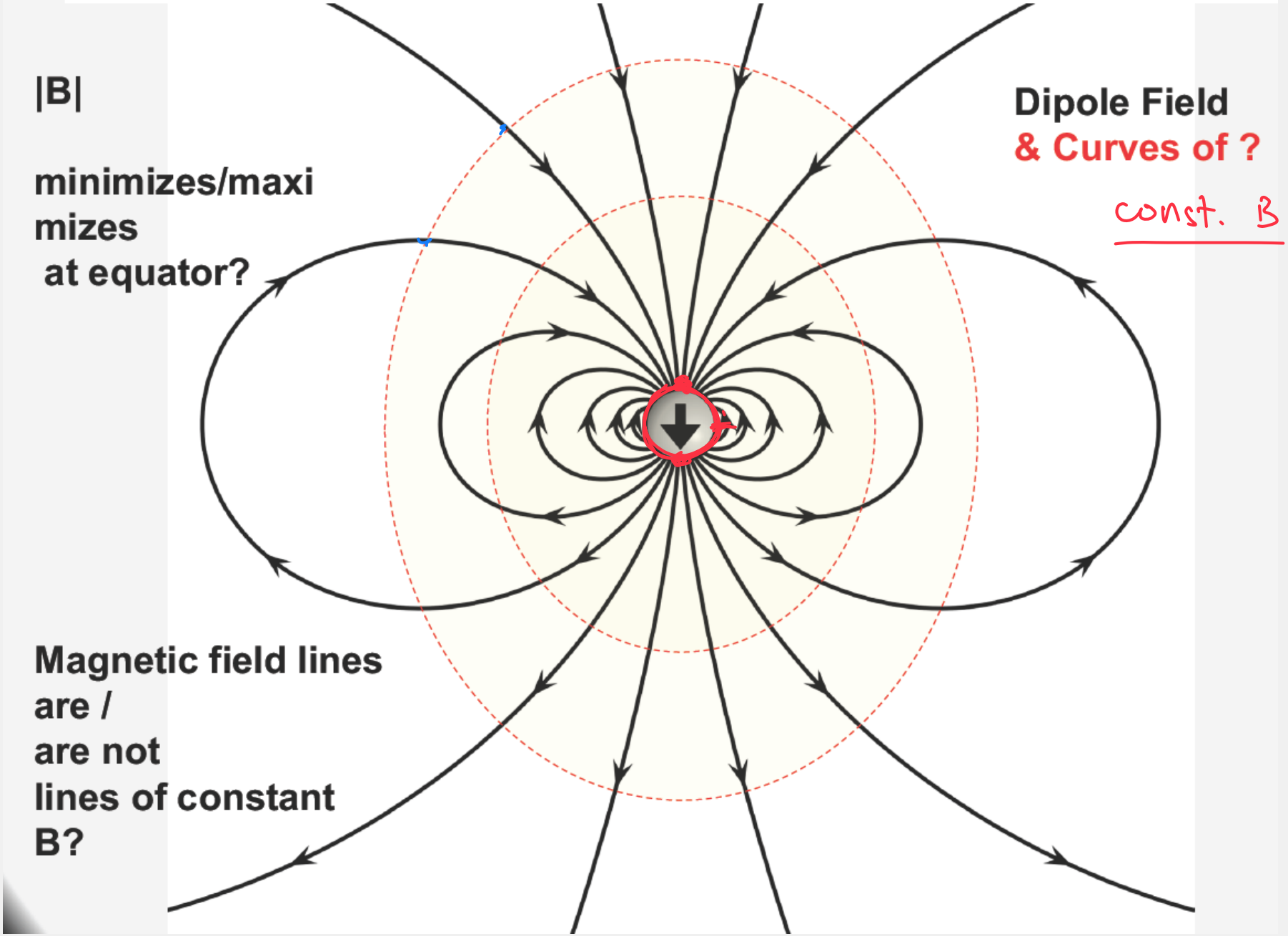
$$\hookrightarrow \quad r = r_0 \cos^2\lambda \implies r = L R_E \cos^2\lambda$$

$L R_E$ → L-shell

at equator: $L = \frac{r}{R_E}$



B-field concepts

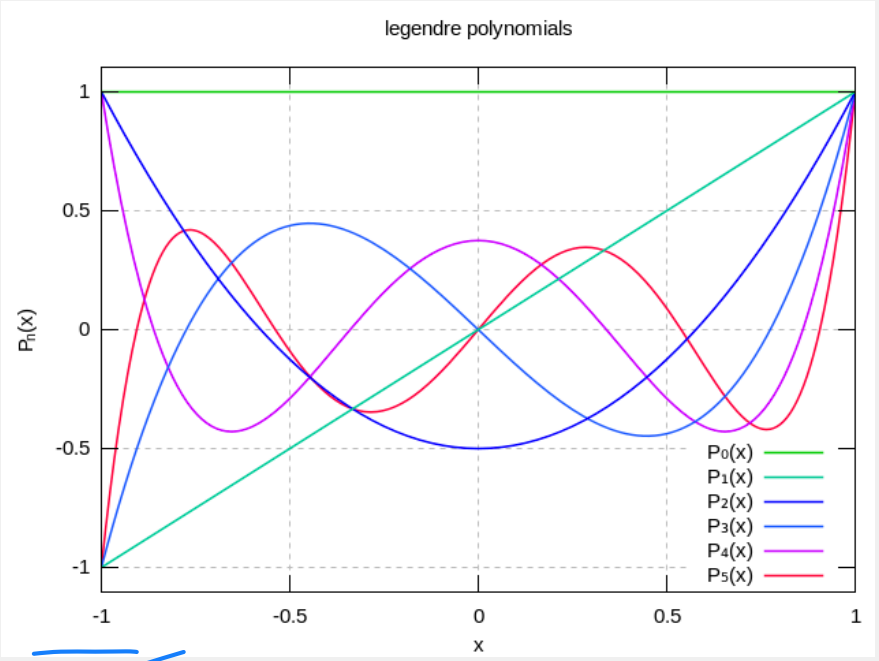


About that extra 10%

- The dipole field model (tilted and off-centered) describes ~90% of the Earth's magnetic field
- For the other 10%, we require a more complex model.
- **Spherical harmonic expansion**
  - Analogous to *Fourier series expansion*, but rather than sines and cosines, uses special functions defined on the surface of a sphere: *Legendre polynomials*

$$P_n(x) = \frac{1}{2^n} \sum_{k=0}^{n} \binom{n}{k}^2 (x-1)^{n-k} (x+1)^k$$

$$P_0(x) = 1$$
$$P_1(x) = x$$
$$P_2(x) = \frac{1}{2}(3x^2 - 1)$$
$$P_3(x) = \frac{1}{2}(5x^3 - 3x)$$

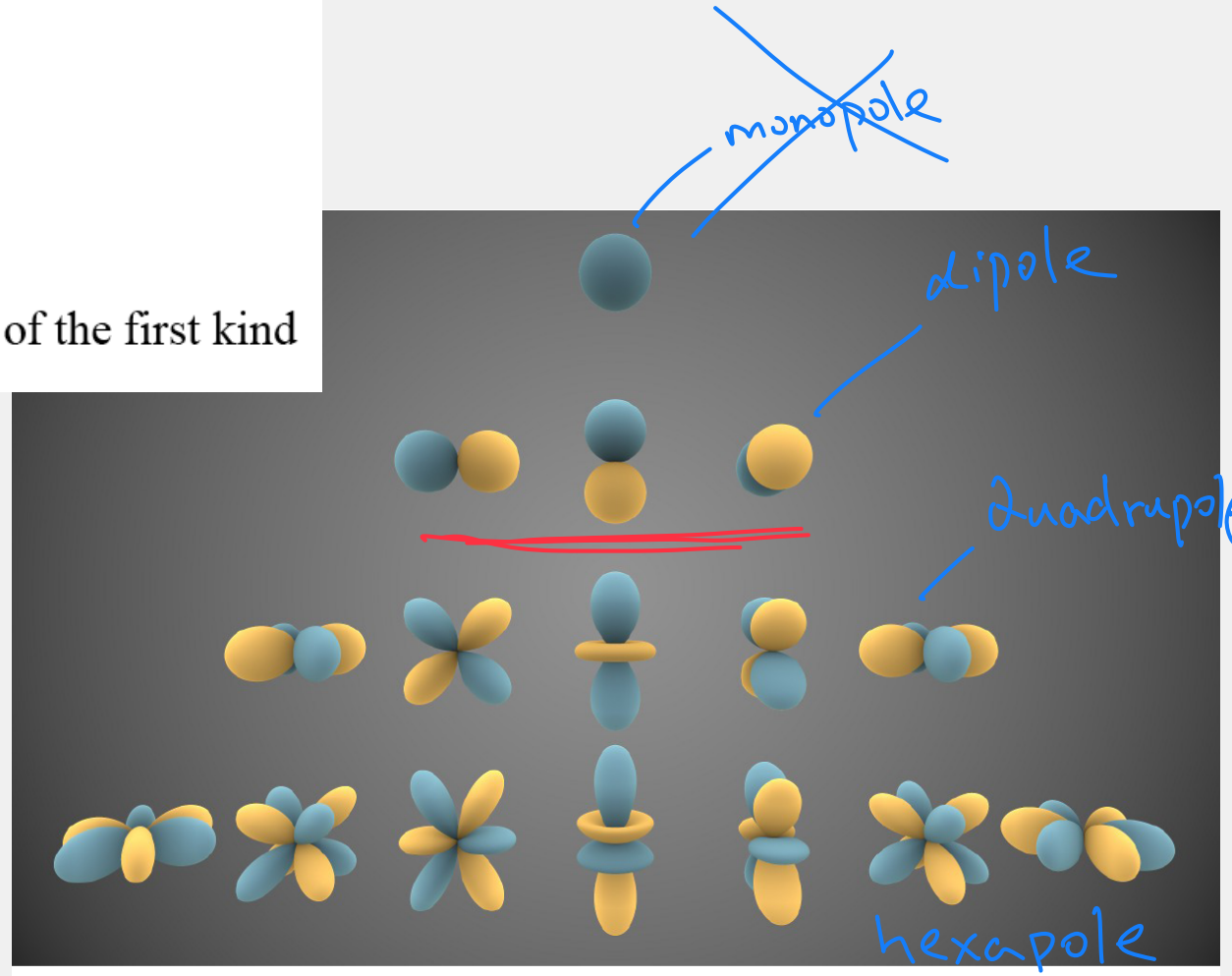


Spherical Harmonic Expansion

$$Y_l^m(\theta,\varphi)=\sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}P_l^m(\cos\theta)\,e^{im\varphi}$$

θ co-latitude
φ longitude
l degree of spherical harmonic
m order of spherical harmonic
$P_l^m(\cos\theta)$ associated Legendre function of the first kind

- Each term (l,m) of the expansion adds a new multipole (in two angle dimensions) to the field



Visual representations of the first few real spherical harmonics. Blue portions represent regions where the function is positive, and yellow portions represent where it is negative. The distance of the surface from the origin indicates the value of $Y_l^m(\theta,\varphi)$ in angular direction (θ,φ).

Earth's Magnetic Field

- In the spherical harmonic expansion, Earth's (internal) B-field is given by:

$$B_{r_i}(r,\theta,\varphi,t) = -\frac{\partial U_i}{\partial r} = \sum_{n=1}^{n_{max}} \left[(n+1) \left(\frac{R_E}{r} \right)^{n+2} \sum_{m=0}^{n} \left(g_n^m(t) \cos m\varphi + h_n^m(t) \sin m\varphi \right) SP_n^m(\cos\theta) \right]$$

$$B_{\theta_i}(r,\theta,\varphi,t) = -\frac{1}{r} \frac{\partial U_i}{\partial \theta} = -\sum_{n=1}^{n_{max}} \left[\left(\frac{R_E}{r} \right)^{n+2} \sum_{m=0}^{n} \left(g_n^m(t) \cos m\varphi + h_n^m(t) \sin m\varphi \right) \frac{dSP_n^m(\cos\theta)}{d\theta} \right]$$

$$B_{\varphi_i}(r,\theta,\varphi,t) = -\frac{1}{r \sin\theta} \frac{\partial U_i}{\partial \varphi} = -\sum_{n=1}^{n_{max}} \left[\left(\frac{R_E}{r} \right)^{n+2} \sum_{m=0}^{n} \left(-g_n^m(t) \sin m\varphi + h_n^m(t) \cos m\varphi \right) \frac{mSP_n^m(\cos\theta)}{\sin\theta} \right]$$

- The "order" of the field model is given by the summation limit n_{max}
- The goal is to find the coefficients g and h
  - *there are* $N = n_{max}^2 + 2\,n_{max}$ *coefficients total!*
  - Typically $n_{max} \sim 14$
- Fit to data using least squares method

U = magnetic scalar potential

$\vec{B} = \nabla U$

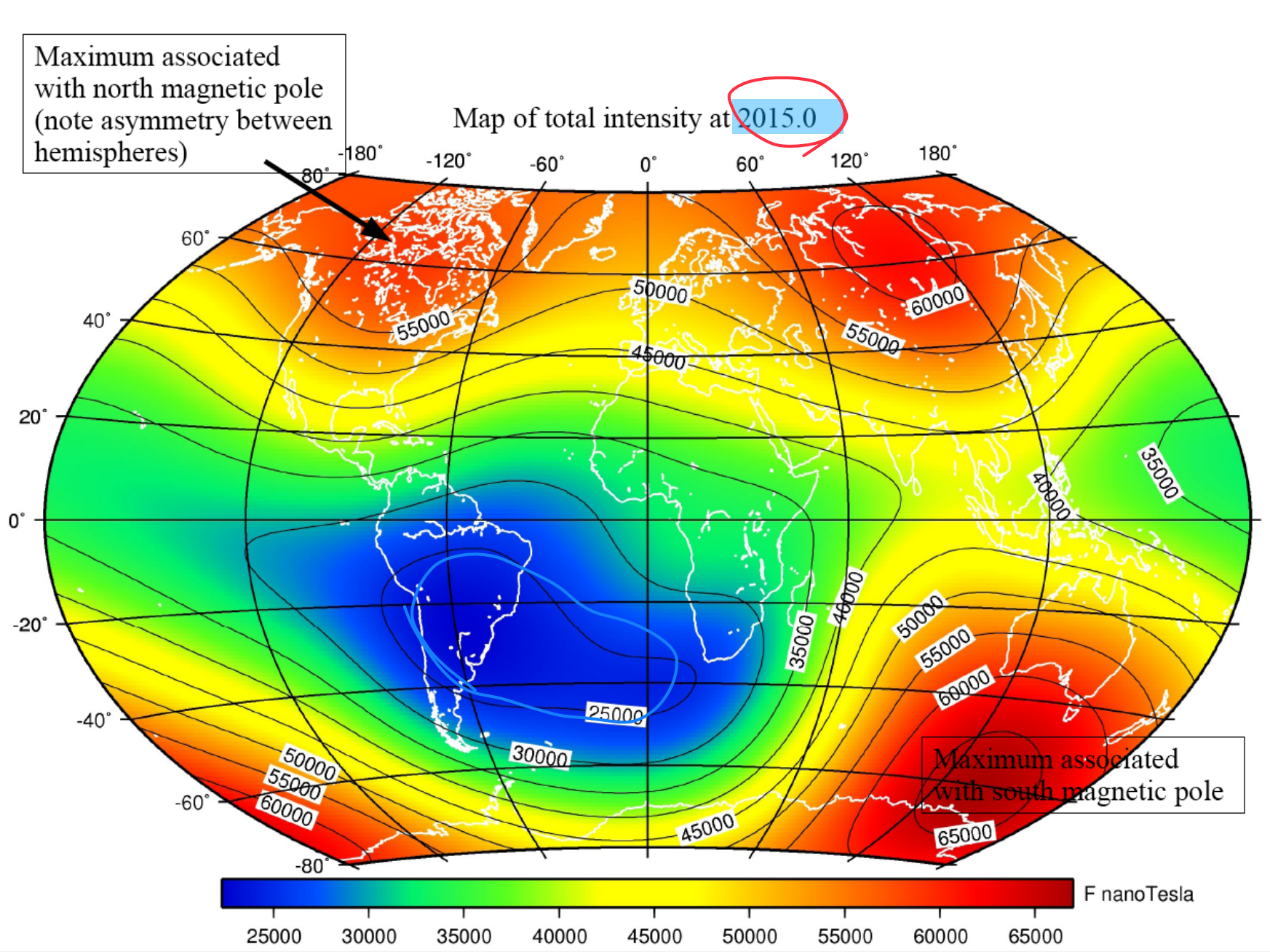
$\nabla \cdot \vec{B} = \nabla \cdot (\nabla U) = 0$

IGRF model

Table 3 12th Generation International Geomagnetic Reference Field

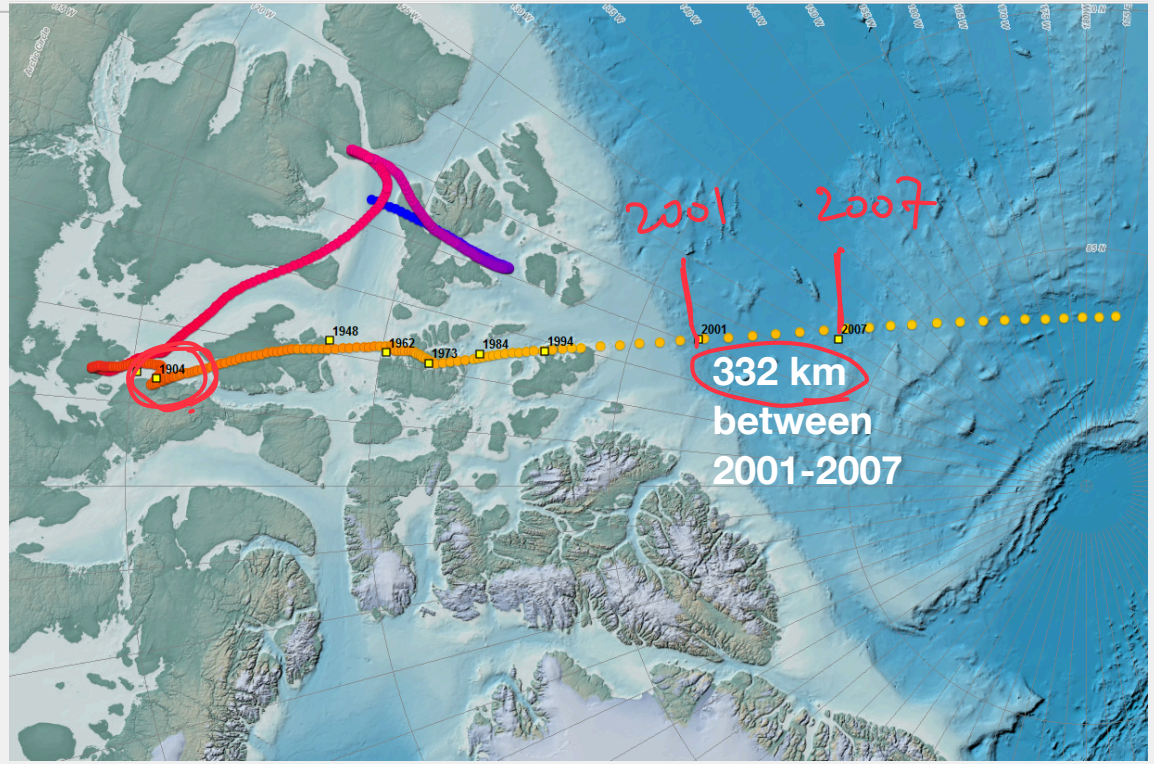
	Degree	Order	IGRF	IGRF	IGRF	IGRF	IGRF	IGRF	IGRF	IGRF	IGRF	DGRF	DGRF	DGRF	DGRF	DGRF	DGRF	DGRF	DGRF	DGRF	DGRF	DGRF	DGRF	IGRF	SV
g/h	n	m	1900.0	1905.0	1910.0	1915.0	1920.0	1925.0	1930.0	1935.0	1940.0	1945.0	1950.0	1955.0	1960.0	1965.0	1970.0	1975.0	1980.0	1985.0	1990.0	1995.0	2000.0	2005.0	2010.0	2015	2015-20
g	1	0	-31543	-31464	-31354	-31212	-31060	-30926	-30805	-30715	-30654	-30594	-30554	-30500	-30421	-30334	-30220	-30100	-29992	-29873	-29775	-29692	-29619.4	-29554.63	-29496.57	-29442.0	10.3
g	1	1	-2298	-2298	-2297	-2306	-2317	-2318	-2316	-2306	-2292	-2285	-2250	-2215	-2169	-2119	-2068	-2013	-1956	-1905	-1848	-1784	-1728.2	-1669.05	-1586.42	-1501.0	18.1
h	1	1	5922	5909	5898	5875	5845	5817	5808	5812	5821	5810	5815	5820	5791	5776	5737	5675	5604	5500	5406	5306	5186.1	5077.99	4944.26	4797.1	-26.6
g	2	0	-677	-728	-769	-802	-839	-893	-951	-1018	-1106	-1244	-1341	-1440	-1555	-1662	-1781	-1902	-1997	-2072	-2131	-2200	-2267.7	-2337.24	-2396.06	-2445.1	-8.7
g	2	1	2905	2928	2948	2956	2959	2969	2980	2984	2981	2990	2998	3003	3002	2997	3000	3010	3027	3044	3059	3070	3068.4	3047.69	3026.34	3012.9	-3.3
h	2	1	-1061	-1086	-1128	-1191	-1259	-1334	-1424	-1520	-1614	-1702	-1810	-1898	-1967	-2016	-2047	-2067	-2129	-2197	-2279	-2366	-2481.6	-2594.50	-2708.54	-2845.6	-27.4
g	2	2	924	1041	1176	1309	1407	1471	1517	1550	1566	1578	1576	1581	1590	1594	1611	1632	1663	1687	1686	1681	1670.9	1657.76	1668.17	1676.7	2.1
h	2	2	1121	1065	1000	917	823	728	644	586	528	477	381	291	206	114	25	-68	-200	-306	-373	-413	-458.0	-515.43	-575.73	-641.9	-14.1
g	3	0	1022	1037	1058	1084	1111	1140	1172	1206	1240	1282	1297	1302	1302	1297	1287	1276	1281	1296	1314	1335	1339.6	1336.30	1339.85	1350.7	3.4
g	3	1	-1469	-1494	-1524	-1559	-1600	-1645	-1692	-1740	-1790	-1834	-1889	-1944	-1992	-2038	-2091	-2144	-2180	-2208	-2239	-2267	-2288.0	-2305.83	-2326.54	-2352.3	-5.5
h	3	1	-330	-357	-389	-421	-445	-462	-480	-494	-499	-499	-476	-462	-414	-404	-366	-333	-336	-310	-284	-262	-227.6	-198.86	-160.40	-115.3	8.2
g	3	2	1256	1239	1223	1212	1205	1202	1205	1215	1232	1255	1274	1288	1289	1292	1278	1260	1251	1247	1248	1249	1252.1	1246.39	1232.10	1225.6	-0.7
h	3	2	3	34	62	84	103	119	133	146	163	186	206	216	224	240	251	262	271	284	293	302	293.4	269.72	251.75	244.9	-0.4
g	3	3	572	635	705	778	839	881	907	918	916	913	896	882	878	856	838	830	833	829	802	759	714.5	672.51	633.73	582.0	-10.1
h	3	3	523	480	425	360	293	229	166	101	43	-11	-46	-83	-130	-165	-196	-223	-252	-297	-352	-427	-491.1	-524.72	-537.03	-538.4	1.8
g	4	0	876	880	884	887	889	891	896	903	914	944	954	958	957	957	952	946	938	936	939	940	932.3	920.55	912.66	907.6	-0.7
g	4	1	628	643	660	678	695	711	727	744	762	776	792	796	800	804	800	791	782	780	780	780	786.8	797.96	808.97	813.7	0.2
h	4	1	195	203	211	218	220	216	205	188	169	144	136	133	135	148	167	191	212	232	247	262	272.6	282.07	286.48	283.3	-1.3
g	4	2	660	653	644	631	616	601	584	565	550	544	528	510	504	479	461	438	398	361	325	290	250.0	210.65	166.58	120.4	-9.1
h	4	2	-69	-77	-90	-109	-134	-163	-195	-226	-252	-276	-278	-274	-278	-269	-266	-265	-257	-249	-240	-236	-231.9	-225.23	-211.03	-188.7	5.3
g	4	3	-361	-380	-400	-416	-424	-426	-422	-415	-405	-421	-408	-397	-394	-390	-395	-405	-419	-424	-423	-418	-403.0	-379.86	-356.83	-334.9	4.1
h	4	3	-210	-201	-189	-173	-153	-130	-109	-90	-72	-55	-37	-23	3	13	26	39	53	69	84	97	119.8	145.15	164.46	180.9	2.9
g	4	4	134	146	160	178	199	217	234	249	265	304	303	290	269	252	234	216	199	170	141	122	111.3	100.00	89.40	70.4	-4.3
h	4	4	-75	-65	-55	-51	-57	-70	-90	-114	-141	-178	-210	-230	-255	-269	-279	-288	-297	-297	-299	-306	-303.8	-305.36	-309.72	-329.5	-5.2
g	5	0	-184	-192	-201	-211	-221	-230	-237	-241	-241	-253	-240	-229	-222	-219	-216	-218	-218	-214	-214	-214	-218.8	-227.00	-230.87	-232.6	-0.2
g	5	1	328	328	327	327	326	326	327	329	334	346	349	360	362	358	359	356	357	355	353	352	351.4	354.41	357.29	360.1	0.5
h	5	1	-210	-193	-172	-148	-122	-96	-72	-51	-33	-12	3	15	16	19	26	31	46	47	46	46	43.8	42.72	44.58	47.3	0.6
g	5	2	264	259	253	245	236	226	218	211	208	194	211	230	242	254	262	264	261	253	245	235	222.3	208.95	200.26	192.4	-1.3
h	5	2	53	56	57	58	58	58	60	64	71	95	103	110	125	128	139	148	150	150	154	165	171.9	180.25	189.01	197.0	1.7
g	5	3	5	-1	-9	-16	-23	-28	-32	-33	-33	-20	-20	-23	-26	-31	-42	-59	-74	-93	-109	-118	-130.4	-136.54	-141.05	-140.9	-0.1
h	5	3	-33	-32	-33	-34	-38	-44	-53	-64	-75	-67	-87	-98	-117	-126	-139	-152	-151	-154	-153	-143	-133.1	-123.45	-118.06	-119.3	-1.2
g	5	4	-86	-93	-102	-111	-119	-125	-131	-136	-141	-142	-147	-152	-156	-157	-160	-159	-162	-164	-165	-166	-168.6	-168.05	-163.17	-157.5	1.4
h	5	4	-134	-125	-126	-126	-125	-122	-118	-115	-113	-119	-122	-121	-114	-97	-91	-83	-78	-75	-69	-55	-39.3	-19.57	-0.01	16.0	3.4
g	5	5	-16	-26	-38	-51	-62	-69	-74	-76	-76	-82	-76	-69	-63	-62	-56	-49	-48	-46	-36	-17	-12.9	-13.55	-8.03	4.1	3.9
h	5	5	3	11	21	32	43	51	58	64	69	82	80	78	81	81	83	88	92	95	97	107	106.3	103.85	101.04	100.2	0.0
g	6	0	63	62	62	61	61	61	60	59	57	59	54	47	46	45	43	45	48	53	61	68	72.3	73.60	72.78	70.0	-0.3
g	6	1	61	60	58	57	55	54	53	53	54	57	57	57	58	61	64	66	66	65	65	67	68.2	69.56	68.69	67.7	-0.1
h	6	1	-9	-7	-5	-2	0	3	4	4	4	6	-1	-9	-10	-11	-12	-13	-15	-16	-16	-17	-17.4	-20.33	-20.90	-20.8	0.0
g	6	2	-11	-11	-11	-10	-10	-9	-9	-8	-7	6	4	3	1	8	15	28	42	51	59	68	74.2	76.74	75.92	72.7	-0.7
h	6	2	83	86	89	93	96	99	102	104	105	100	99	96	99	100	100	99	93	88	82	72	63.7	54.75	44.18	33.2	-2.1
g	6	3	-217	-221	-234	-228	-233	-238	-242	-246	-249	-246	-247	-247	-237	-228	-212	-198	-192	-185	-178	-170	-160.9	-151.34	-141.40	-129.9	2.1
h	6	3	2	4	5	8	11	14	19	25	33	16	33	48	60	68	72	75	71	69	69	67	65.1	63.63	61.54	58.9	-0.7
g	6	4	-58	-57	-54	-51	-46	-40	-32	-25	-18	-25	-16	-8	-1	4	2	1	4	4	3	-1	-5.9	-14.58	-22.83	-28.9	-1.2
h	6	4	-35	-32	-29	-26	-22	-18	-16	-15	-15	-9	-12	-16	-20	-32	-37	-41	-43	-48	-52	-58	-61.2	-63.53	-66.26	-66.7	0.2
g	6	5	59	57	54	49	44	39	32	25	18	21	12	7	-2	1	3	6	14	16	18	19	16.9	14.58	13.10	13.2	0.3
h	6	5	36	32	28	23	18	13	8	4	0	-16	-12	-12	-11	-8	-6	-4	-2	-1	1	1	0.7	0.24	3.02	7.3	0.9
g	6	6	-90	-92	-95	-98	-101	-103	-104	-106	-107	-104	-105	-107	-113	-111	-112	-111	-108	-102	-96	-93	-90.4	-86.36	-78.09	-70.9	1.6
h	6	6	-69	-67	-65	-62	-57	-52	-46	-40	-33	-39	-30	-24	-17	-7	1	11	17	21	24	36	43.8	50.94	55.40	62.6	1.0
g	7	0	70	70	71	72	73	73	74	74	74	70	65	65	67	75	72	71	72	74	77	77	79.0	79.88	80.44	81.6	0.3
g	7	1	-55	-54	-54	-54	-54	-54	-54	-53	-53	-40	-55	-56	-56	-57	-57	-56	-59	-62	-64	-72	-74.0	-74.46	-75.00	-76.1	-0.2
h	7	1	-45	-46	-47	-48	-49	-50	-51	-52	-52	-45	-35	-50	-55	-61	-70	-77	-82	-83	-80	-69	-64.6	-61.14	-57.80	-54.1	0.8
g	7	2	0	0	1	2	2	3	4	4	4	0	2	2	5	4	1	1	2	3	2	1	0.0	-1.65	-4.55	-6.8	-0.5

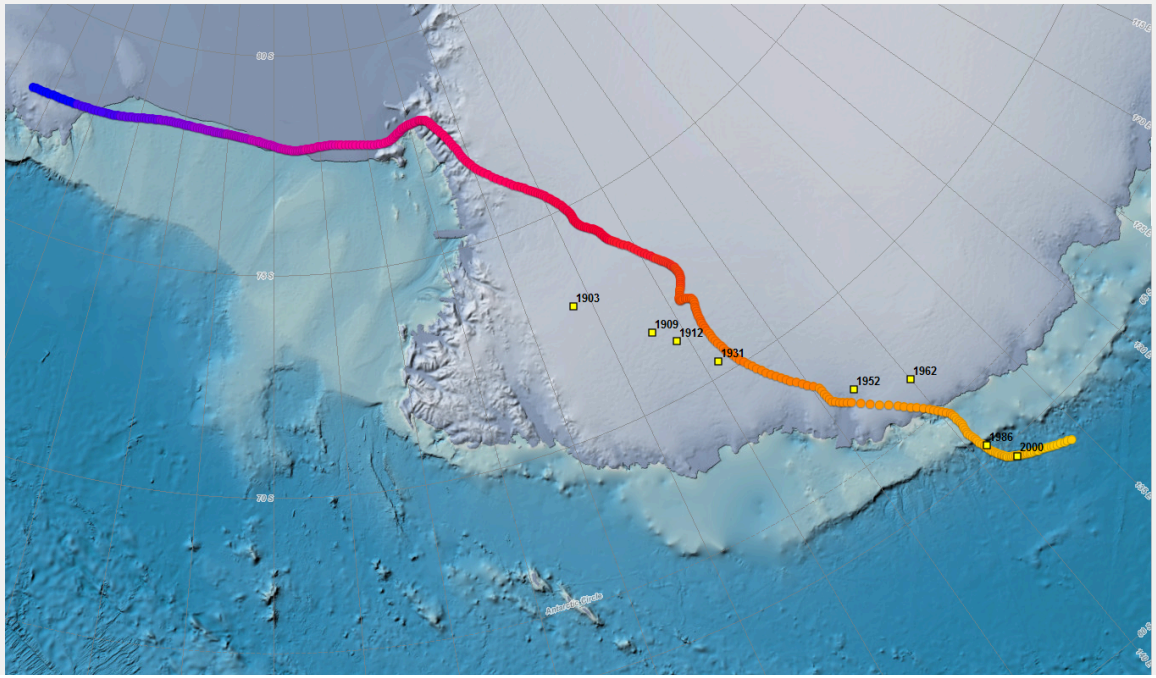
IGRF


Maximum associated with north magnetic pole (note asymmetry between hemispheres)
Map of total intensity at 2015.0
Maximum associated with south magnetic pole
F nanoTesla
25000
30000
35000
40000
45000
50000
55000
60000
65000

The Magnetic Field is Changing… slowly

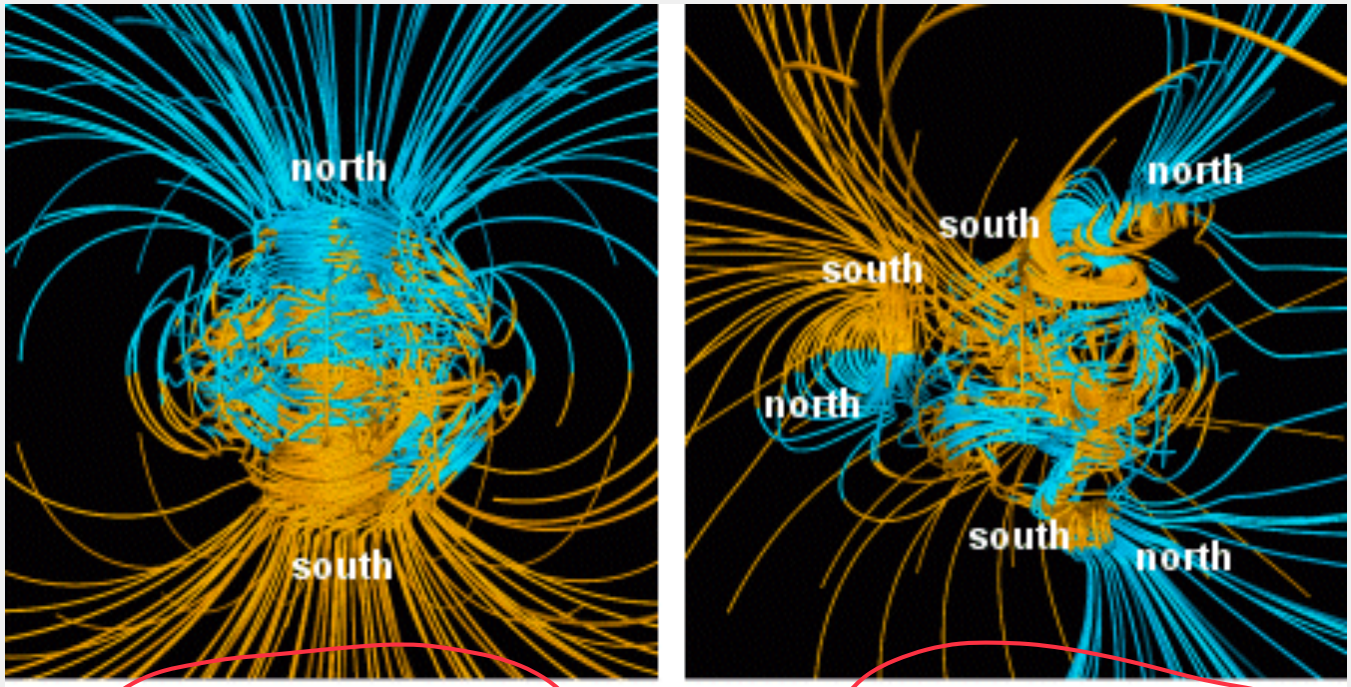
- Magnetic field changes on timescales of years, decades, millennia
- **Magnetic field "flips" about every 500,000 years** (with high variance)
  - Last reversal was 780,000 years ago
  - Measured through magnetic materials and/or volcanic material
- **Change most easily observed as drift of magnetic poles**
  - North pole first measured in 1831; drifting northeast since then
  - South pole first measured in 1903





Flips due to dynamics in the outer core

- The magnetic field of the Earth is generated by **dynamo action** in which **convection** of molten iron in the planetary core generates electric currents which in turn give rise to magnetic fields
- In **simulations** of planetary dynamos, reversals often emerge spontaneously from the underlying dynamics

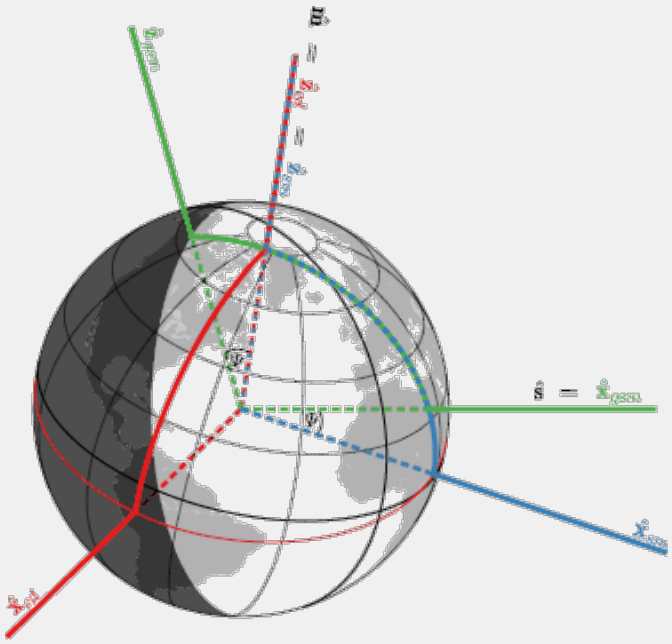


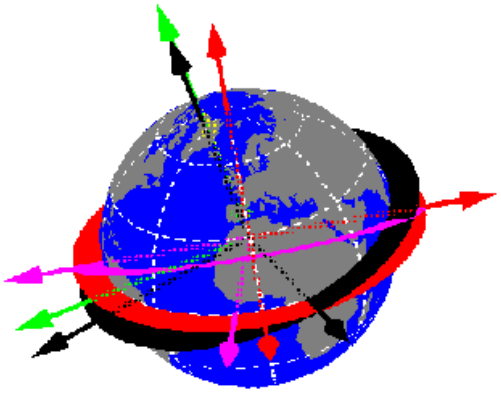
University of Colorado **Boulder**

Magnetosphere: Beyond the Surface

More on Coordinate Systems

- Geographic:
- Geodetic:
- GEI = Geocentric Equatorial Inertial:
- Geomagnetic:
- Geocentric Solar Ecliptic (GSE):
- Geocentric Solar Equatorial (GSEQ):
- Geocentric Solar Magnetospheric (GSM):
- Solar Magnetic (SM):
- Spacecraft:
- Transformation matrices map from one coordinate system to another





Axes

	X	Y	Z
GEO			
GEI			
MAG			
SM			
GSM			

Coordinates

Geodetic:

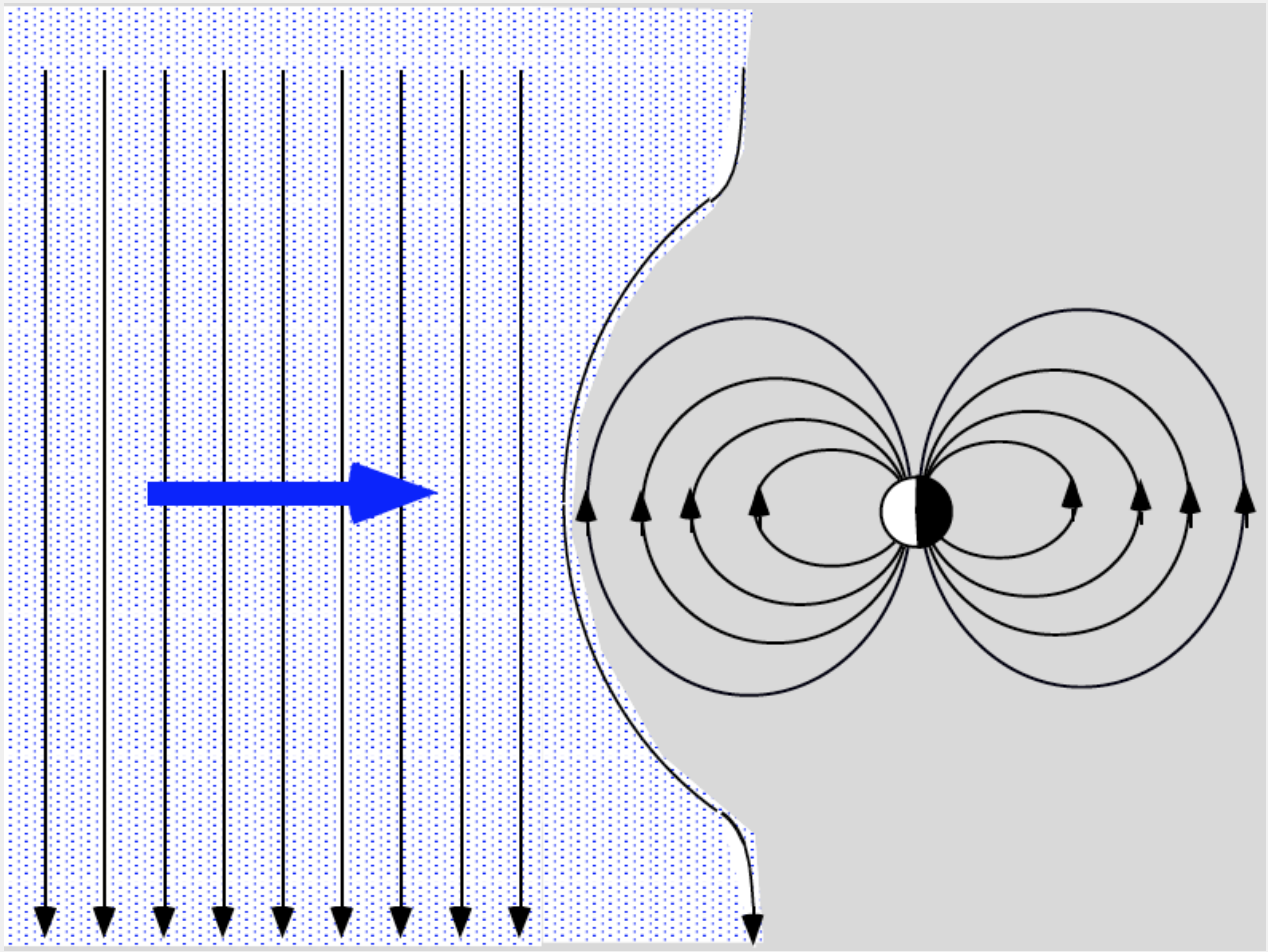
0.1504 50.7978 4.3592

Radius: 6365.5119

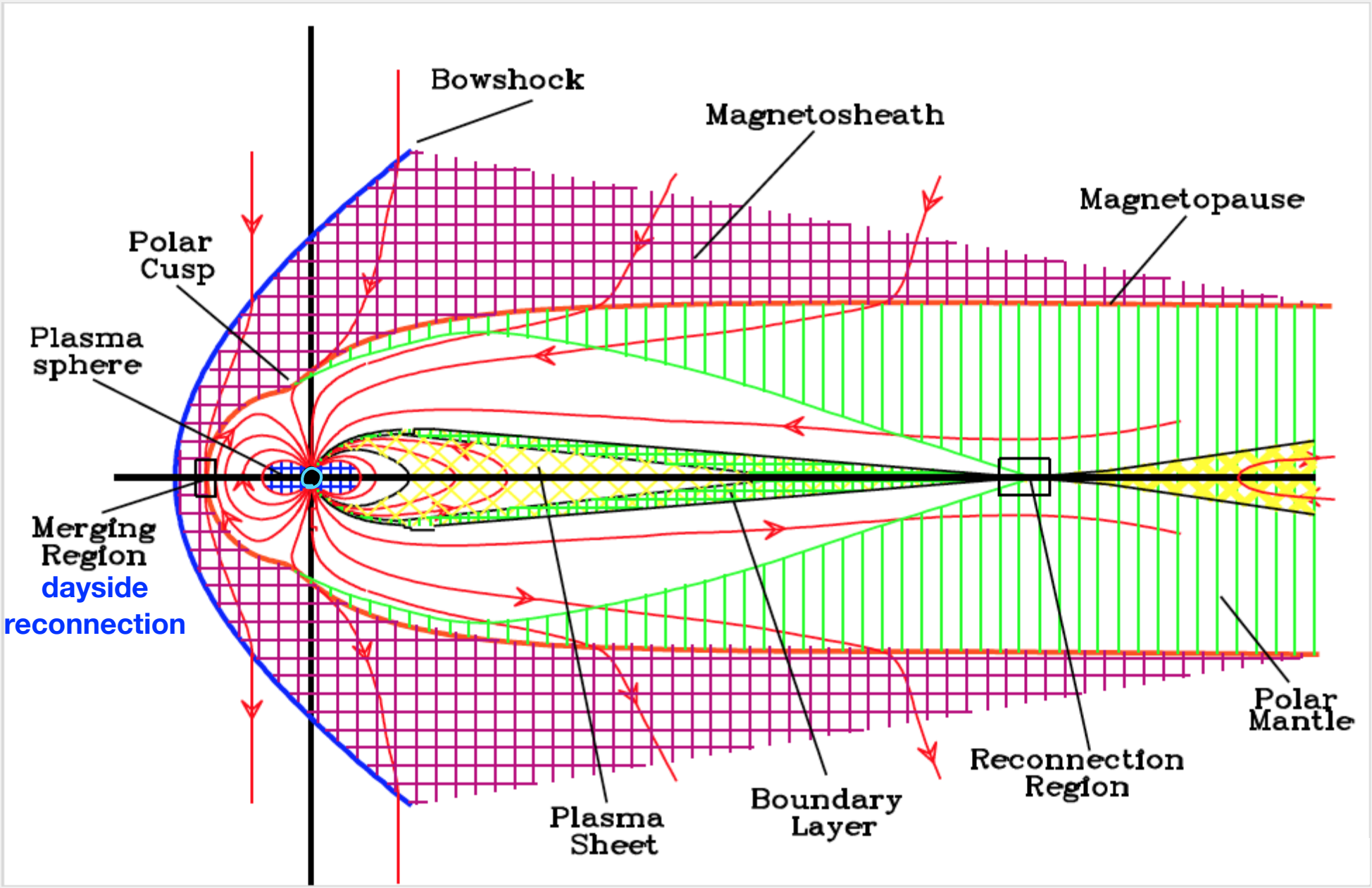
	latitude	longitude
GEO	50.6092	4.3592
GEI	50.6092	104.8044
MAG	51.9947	89.1577
SM	51.9947	−167.9438
GSM	26.5415	−171.7345
GEI_{J2000}	50.6120	104.8169

Magnetic Field in Space

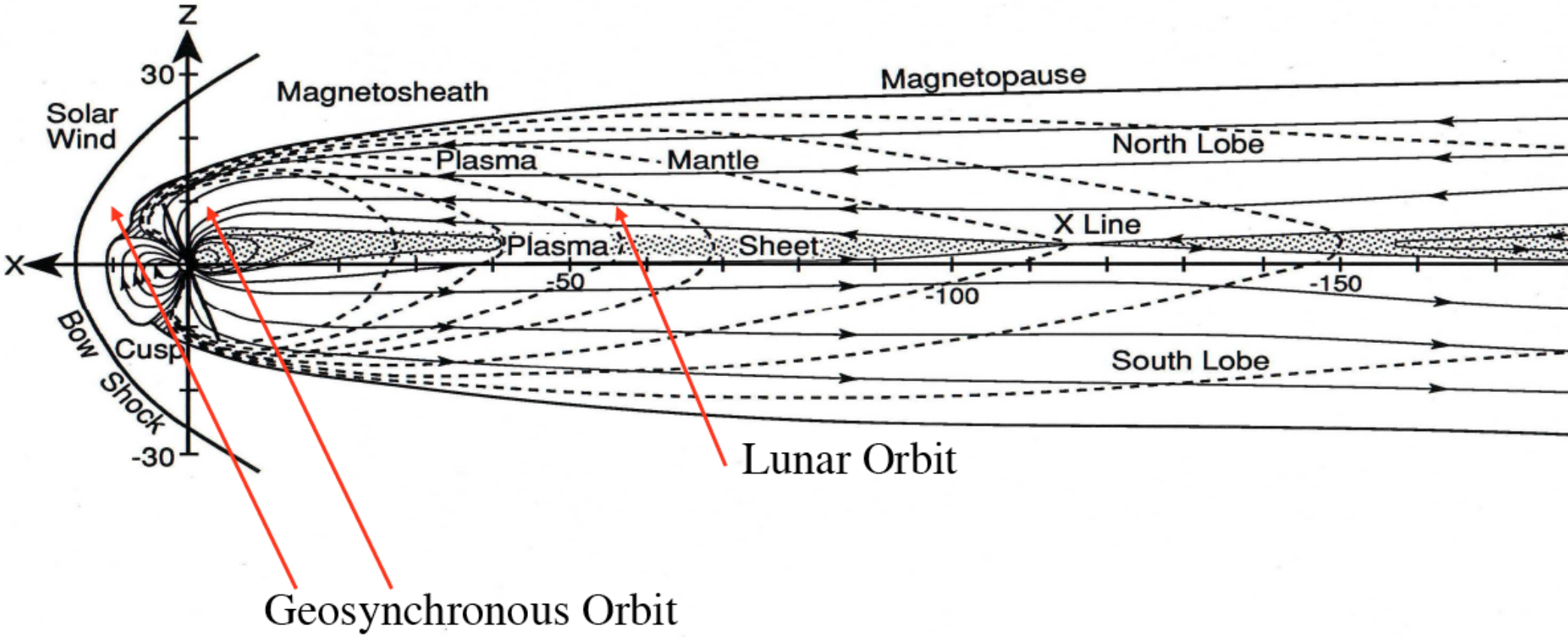
- As we get further out into space, the dipole (multipole) field model breaks down
- Field is heavily distorted by the solar wind
  - Solar wind carries particles (dynamic pressure) and drags the interplanetary magnetic field (IMF)



Magnetosphere

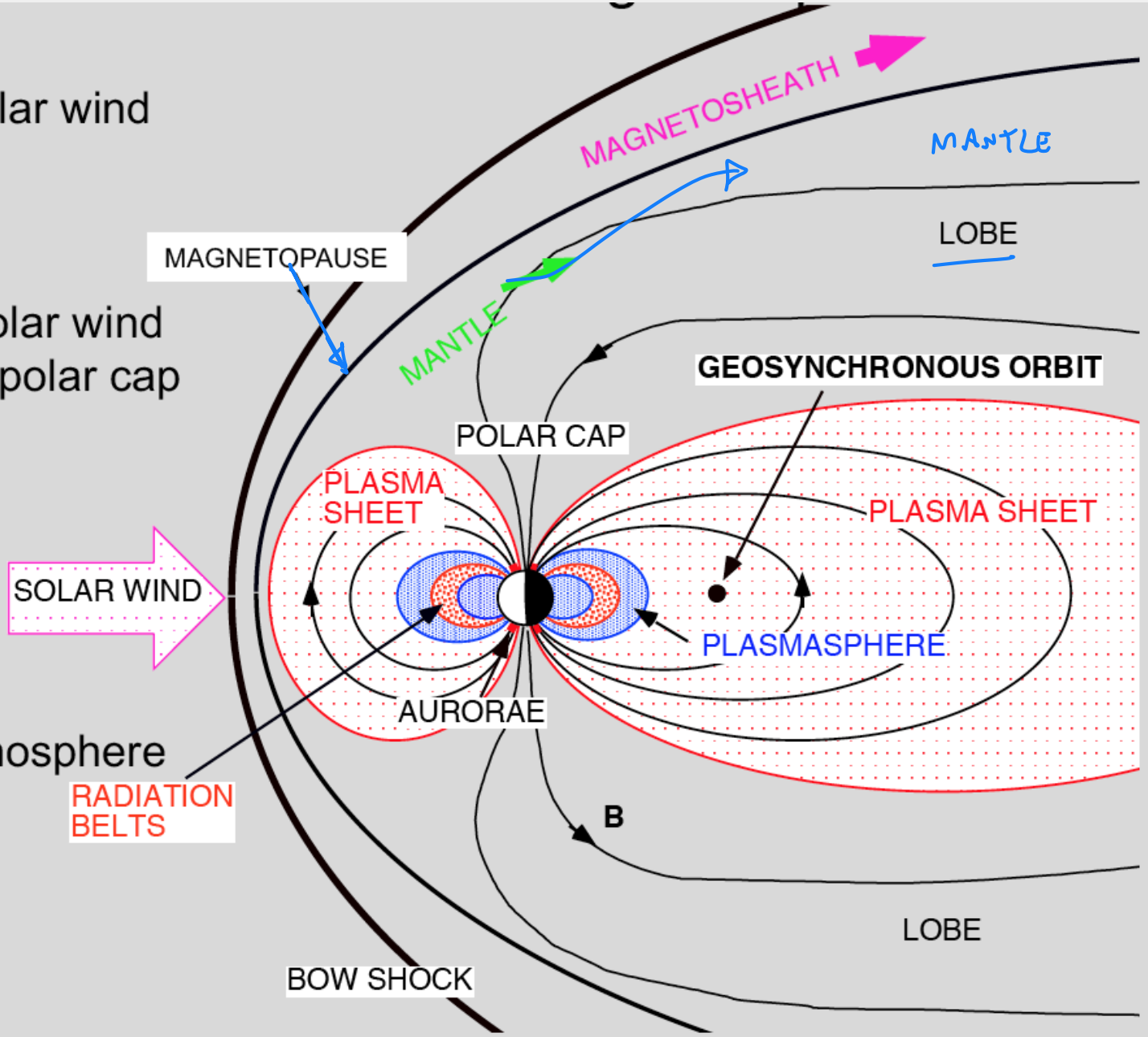


Magnetosphere to Scale



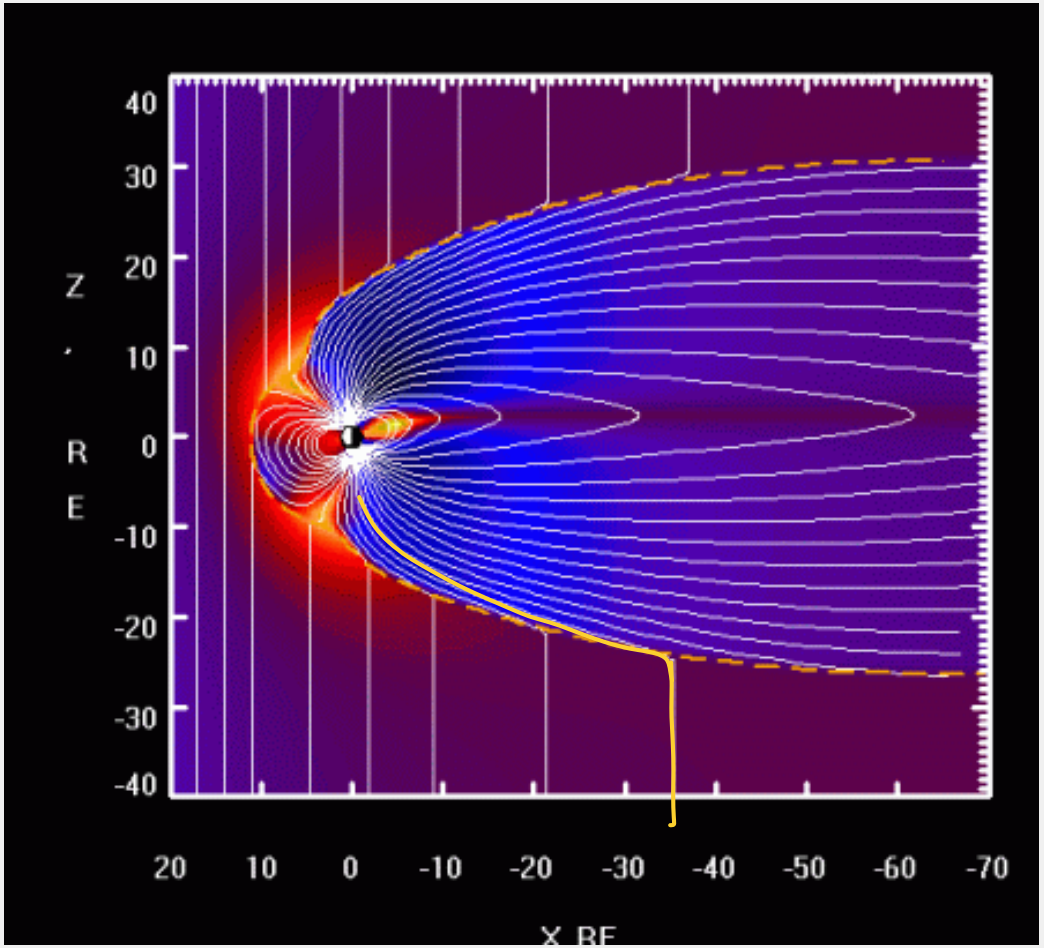
Basic Structure of (Inner) Magnetosphere

- Magnetosheath
  - Compressed solar wind
- Mantle
  - Outer cloak
- Lobe
  - Connected to solar wind downtail and to polar cap at Earth
  - Cool
- Plasmasheet
  - Flexible dipole
  - Warm
- Plasmasphere
  - Extension of Ionosphere
  - Cold
- Radiation Belts
  - Mostly Dipole
  - Hot



There's no way you could possibly model this mess.

- But Nikolai Tsyganenko (University of St. Petersburg) is smarter than me.
- https://ccmc.gsfc.nasa.gov/models/modelinfo.php?model=Tsyganenko%20Magnetic%20Field
- semi-empirical fit to a whole ton of spacecraft magnetometer data: IMP, HEOS, ISEE, POLAR, Geotail, GOES



- Model updated every so often: T89, T96, T01, T02, TS07 (*S is for Sitnov*)
- TS07 continually updated with new data